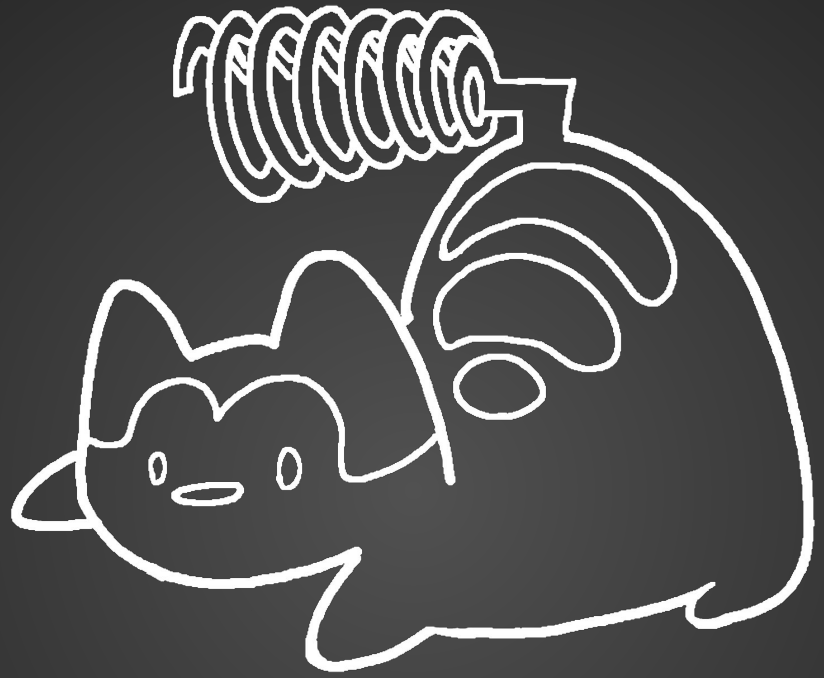




Mesh Networks For Beginners

Hacking with Meshtastic, Meshcore, & Reticulum

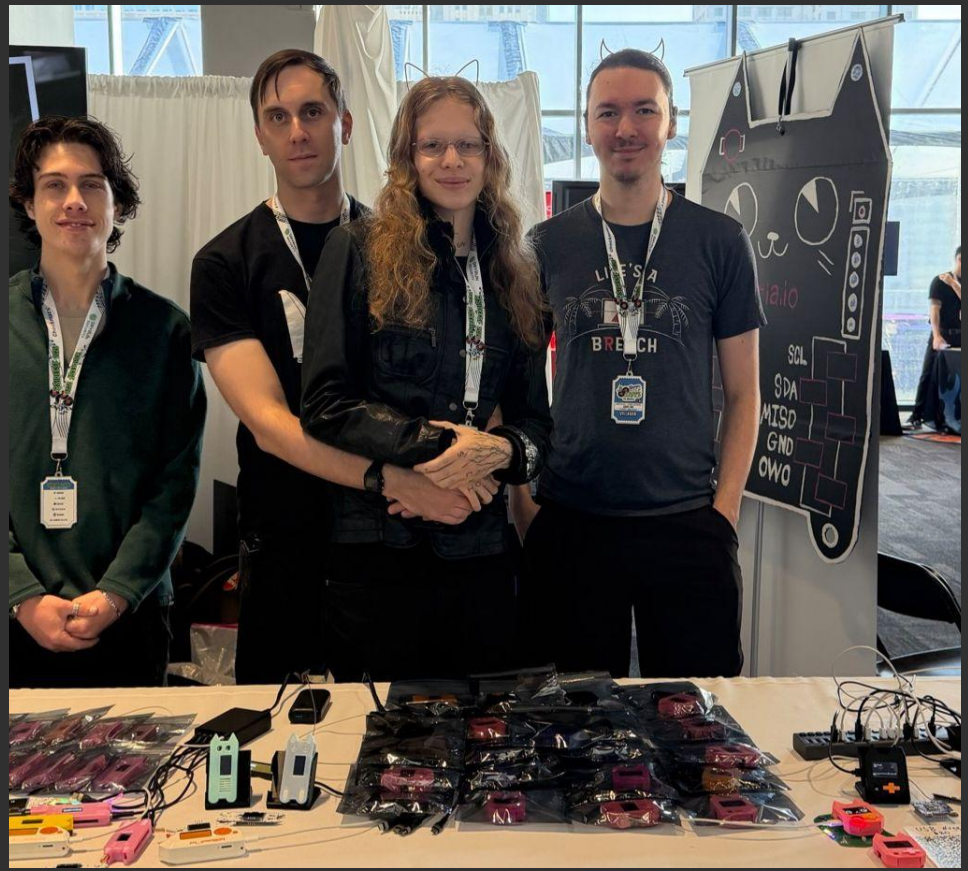
What are we learning today?





- Common & custom types of LoRa Nodes
- Legal & practical considerations
- LoRa & Mesh networking fundamentals
- Meshtastic vs Meshcore
- How to set up new nodes for Meshtastic, Meshcore & Reticulum
- Configuring settings to join local community nets
- **Meshlighter - Make Radio Nerds Very Angry**

Who Am I?



I'm Kody Kinzie!

You can find me at www.hack.gay

Security researcher focused on radio hacking, digital investigations, and cheap electronics.

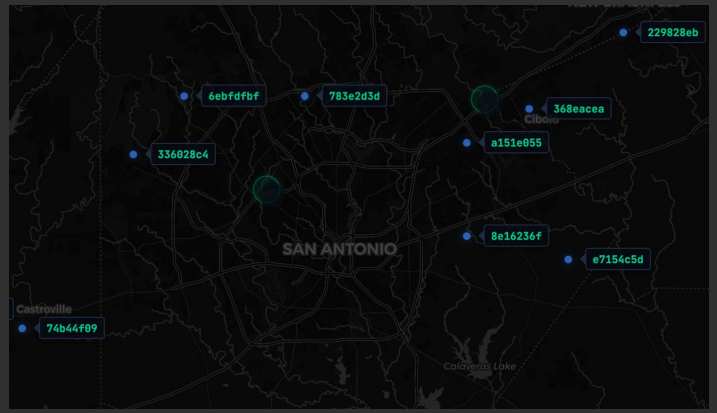
I love teaching beginners how to get started with ethical hacking through:

- Script Kitty village
- Ethical Hacking Cyber camps
- Workshops
- Custom Hardware

You may have seen me on youtube:

Null Byte, Hak5, ScriptKitty

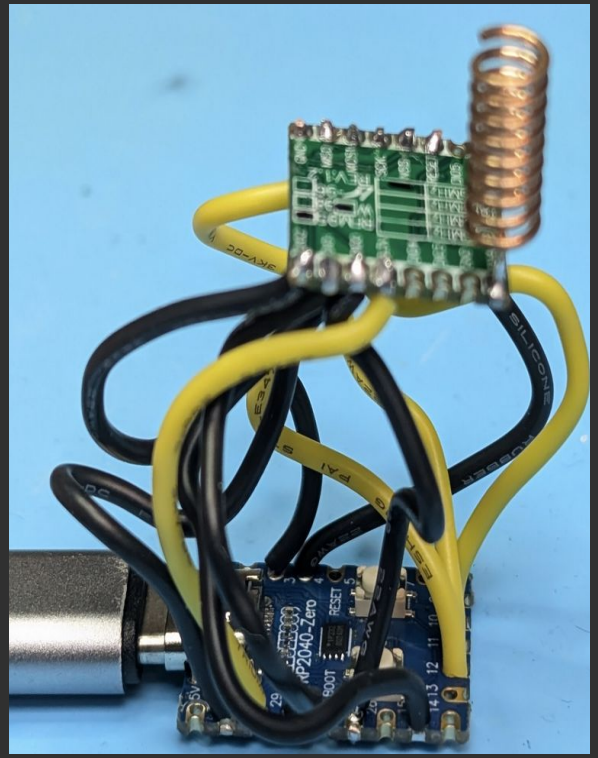
What is the point of Mesh Networks?



- Bidirectional, encrypted off-grid communication
- Ultra-low cost hardware makes nodes disposable
- Monitor sensors, track objects, remote control
- Backup communication in areas with congested or nonexistent infrastructure
- Amplified by nearby nodes, regional networks span 200+ miles

What is the cost of participating?

- You can make your own horrible nodes for $5
- $70-$90 for a fancy node from China
- $120+ for ruggedized, fully featured nodes



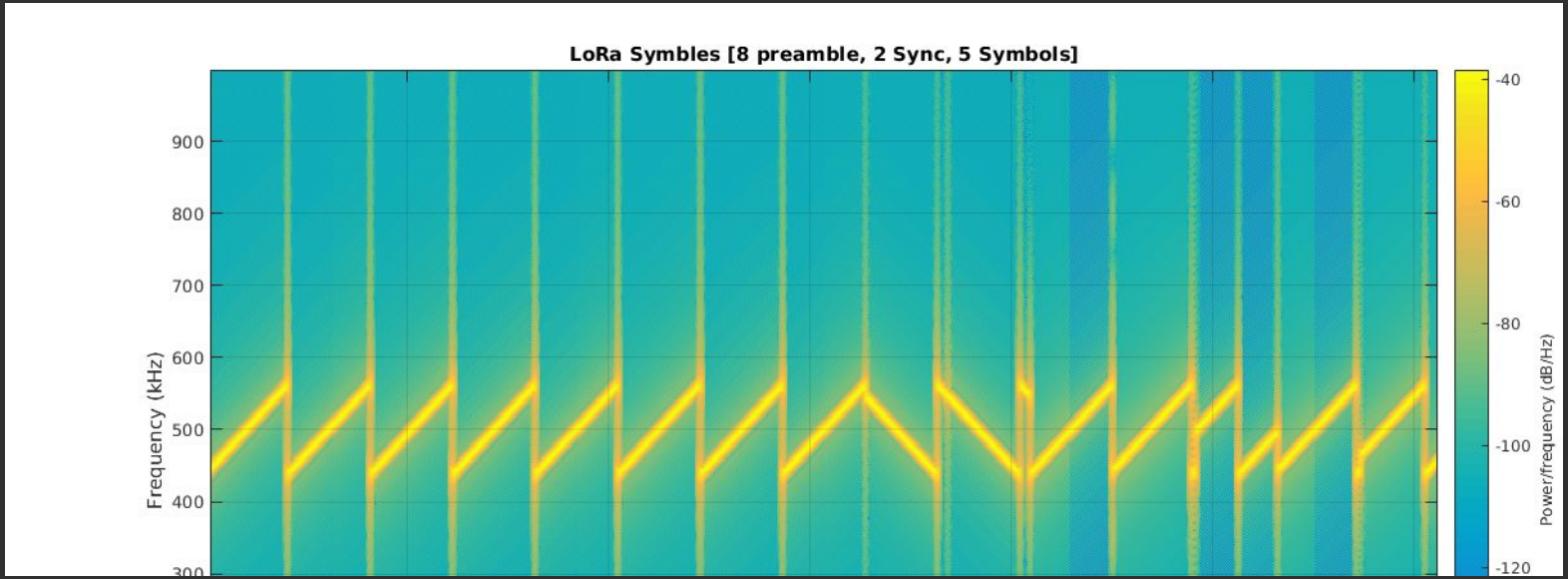
What can we make with this?



- Off-grid chat in backwoors/remote areas
- Chat with congested cellular service
- GPS Tracking of moving nodes
- Map and military intelligence sharing
- Anti-satellite warfare resilience
- Encrypted disaster/emergency response
- Communication between vehicles, boats, & aircraft
- Remote Sensor telemetry (presence, environmental)
- Remote control of anything: https://www.youtube.com/watch?v=hS_IR4AmvD8

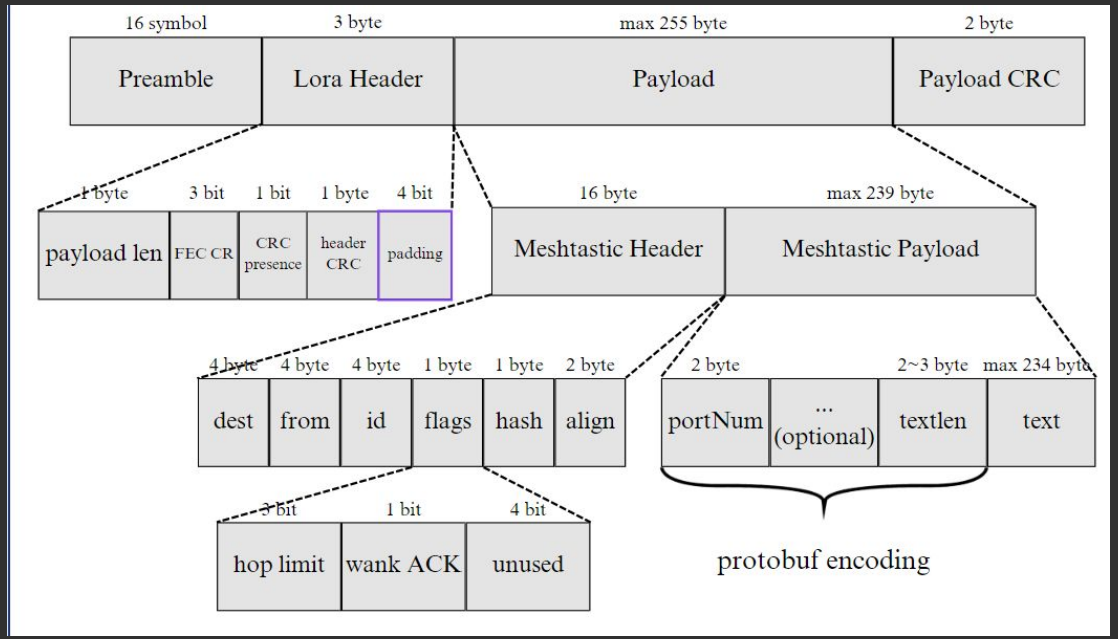
Defining Terms: LoRa Radio

- LoRa - The long range radio standard we use to transmit, encoding data using up and down chirps



Defining Terms: Meshtastic

- Meshtastic - A protocol, with packets sent over LoRa, that allows for encrypted mesh networking primarily at Layer 3 (Network Layer) of the OSI model



What is Meshtastic?

Meshtastic adds encryption, managed flood routing, & convenient apps to LoRa

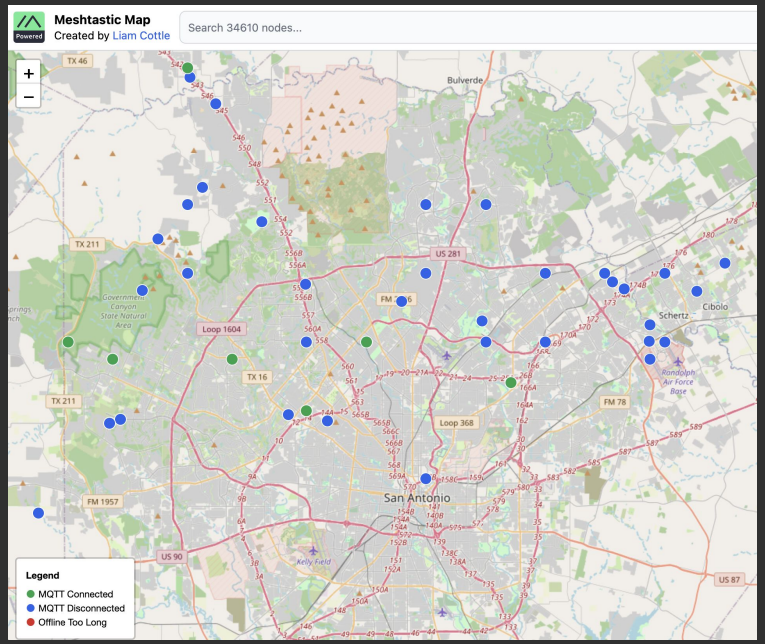
- Users control nodes via Bluetooth, Wi-Fi, or Serial connection
- Managed with iOS, Android, MacOS or Windows apps
- Can also connect via any chrome browser with WebSerial webapp!
- Encrypts packets & removes headaches of matching radio settings

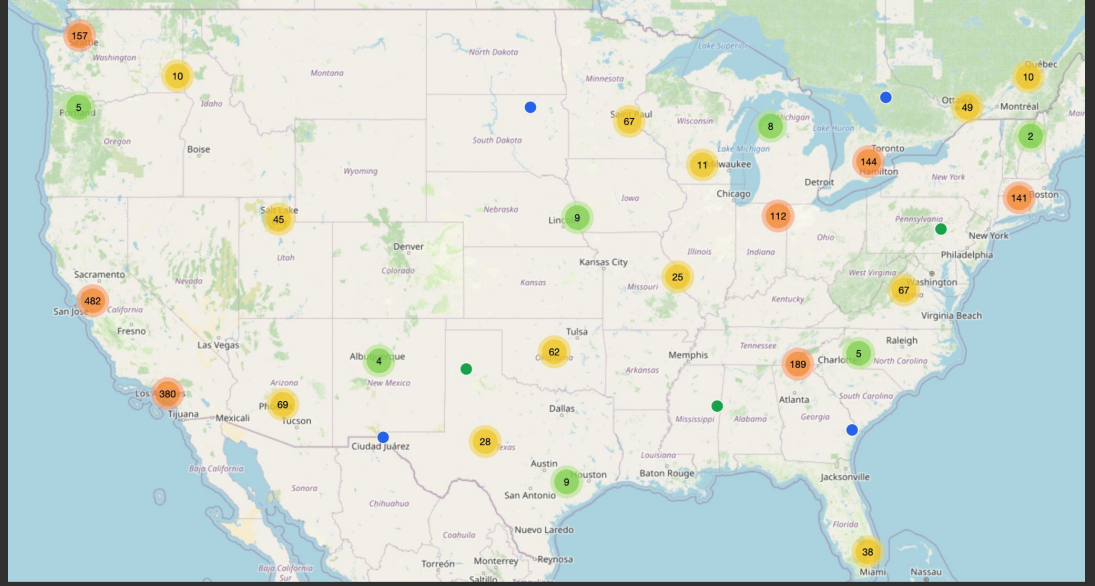


Real World Mapping

meshtastic.liamcottle.net

Uses local MQTT collectors



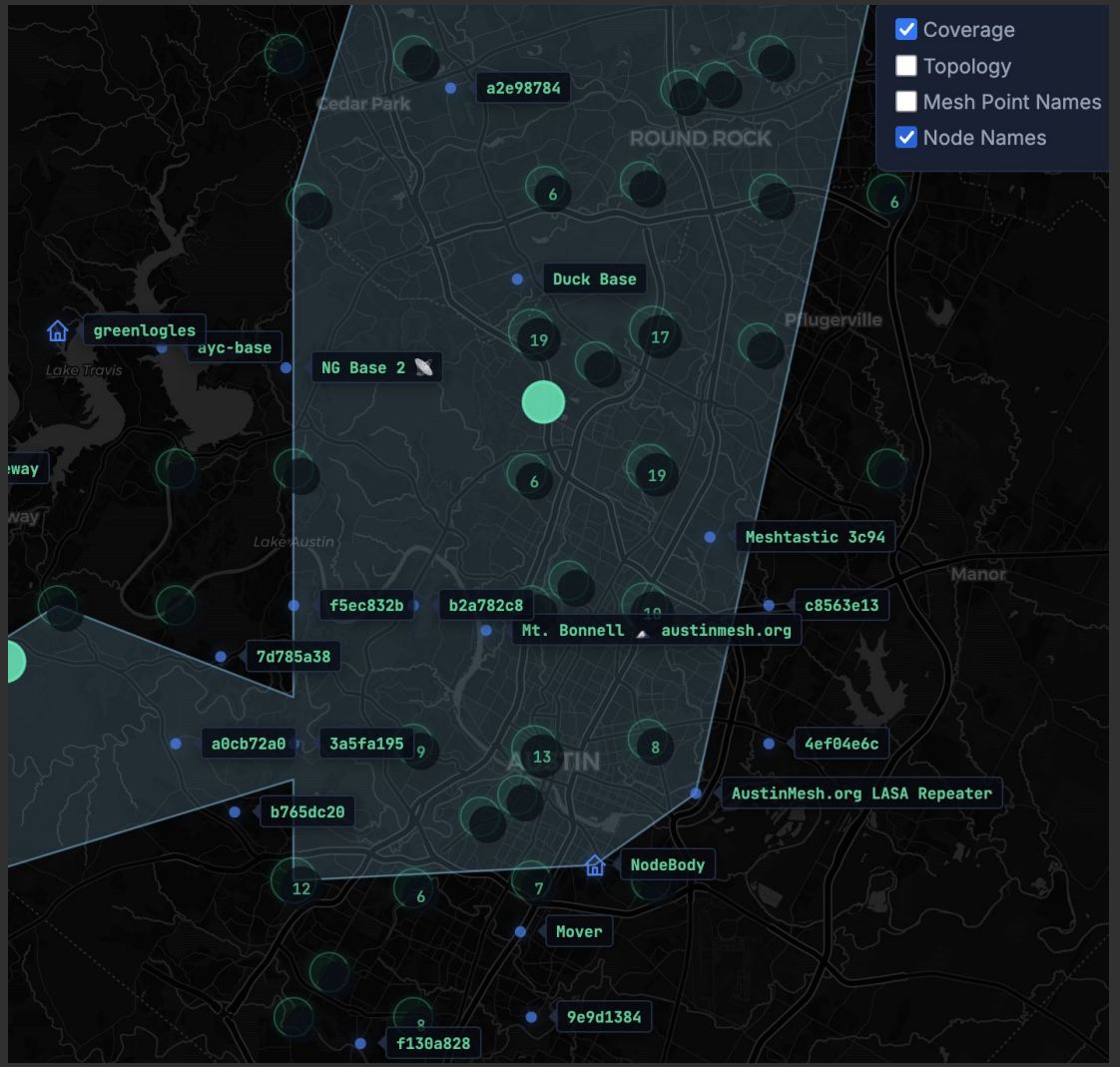


- Collector nodes forward observed nodes over internet to the map via MQTT
- Blind spots, but I can track location of nodes over time
- Tracked my flight!

Also: MeshRadar

https://meshradar.io

Also very useful for finding the settings local mesh nets are using!



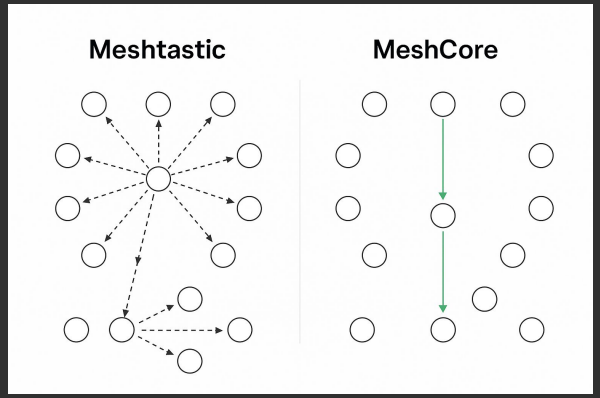
Defining Terms: Meshcore

- Meshcore - Another Layer 3 protocol introduced in 2025, designed to be more scalable and allow for more reliable direct messaging than Meshtastic



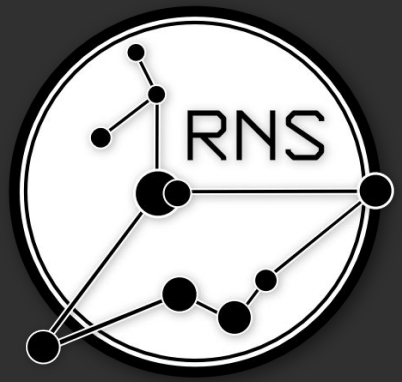
Meshcore Vs Meshtastic - Push vs Pull Telemetry

- Meshtastic uses flood routing, all devices repeat traffic they hear by default
  - This is loud, many devices are transmitting and causes network traffic and congestion
  - Lots of unnecessary traffic in default networks
  - Maximum of 7 hops and no defined routes means low reliability for DM’s
- Meshcore uses static routes, floods only when needed, only repeaters repeat packets they hear
  - Much more quiet because only repeaters send telemetry and repeat
  - Manually send adverts to be discovered
  - Remembering paths prevents unnecessary flooding and scales better



Meshtastic & Meshcore Alternatives

- LoRaWAN - Uses a star topology, where end-devices (sensors) communicate directly with high-power gateways. Devices only talk to the gateway.
- Reticulum - A cryptography-based networking stack for building local and wide-area networks with readily available hardware, including Wi-Fi and LoRa



About MeshCore

MeshCore is a multi platform system for enabling secure text based communications utilising LoRa radio hardware. It can be used for Off-Grid Communication, Emergency Response & Disaster Recovery, Outdoor Activities, Tactical Security including law enforcement, private security and also IoT sensor networks.

Our Mission

At MeshCore, our passion is building reliable and secure decentralised mesh radio networks for text-based communication. We are driven by the need to empower users in off-grid and disaster-stricken regions with a communication tool that does not depend on the internet or cellular networks.

Easy to use

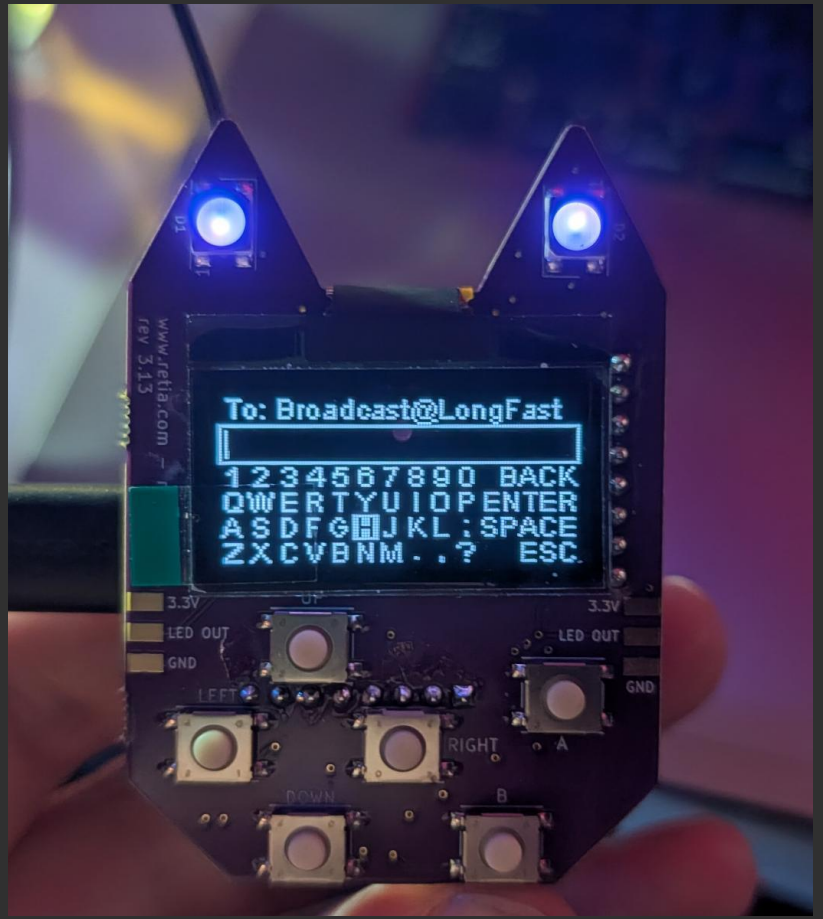
Use our web flasher to install MeshCore firmware on popular devices like the Lilygo T-Deck or Heltec V3 LoRa32, you can also use our web client to communicate securely over radio with other users just using a Chrome web browser - and yes of course it works without the internet!

Hardware Fundamentals

Selecting or making a node starts with understanding how they work!

I’ve been designing Meshtastic nodes for 3 years now

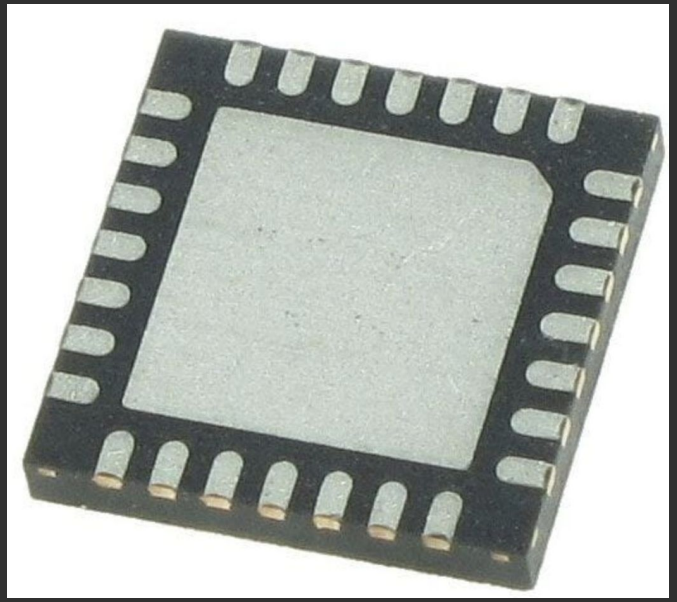
Here is what I’ve learned:



Defining Terms: LoRa Radio Chips

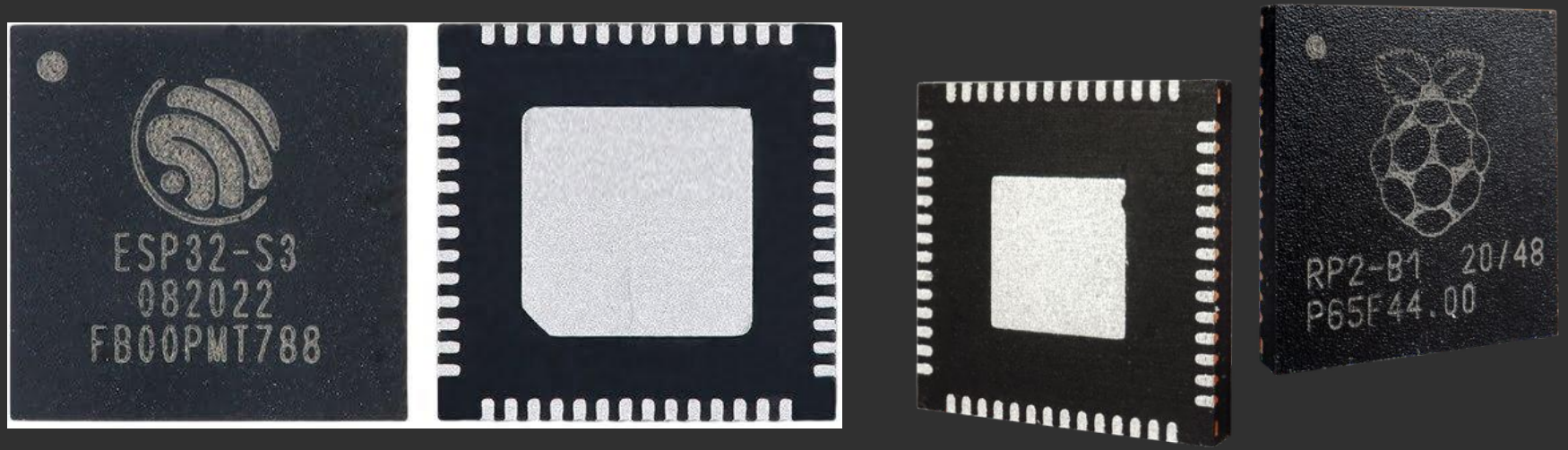
- **LoRa Chipset** - The LoRa radio chip that controls radio transmission, such as sx1262. Pick careful! Some are locked to a frequency.





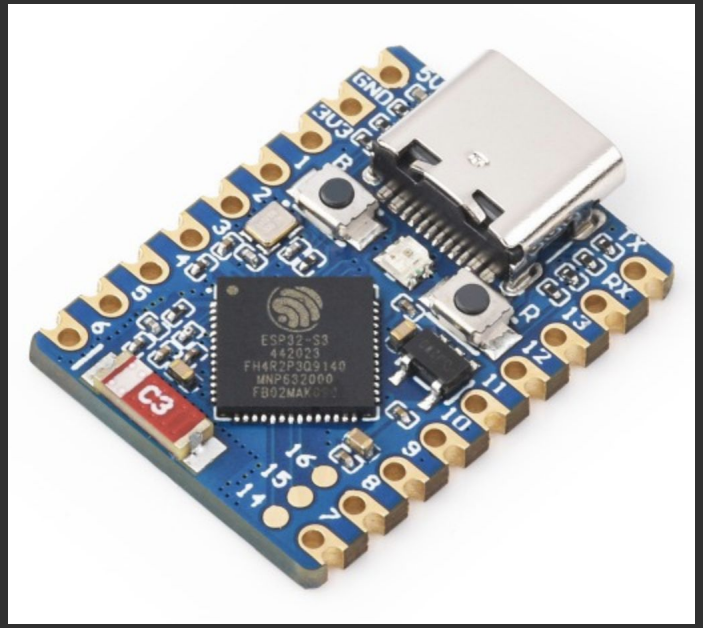
Defining Terms: Host Microcontroller

- **Host Microcontroller** - The “brains” that run Meshtastic, handle Bluetooth, Wi-Fi, & Serial communication, & interface with external hardware



Defining Terms: Breakout Boards

- **Breakout board**: An expansion board designed to make a bare host or radio chip more useful, breaking out pins & sometimes USB & power management

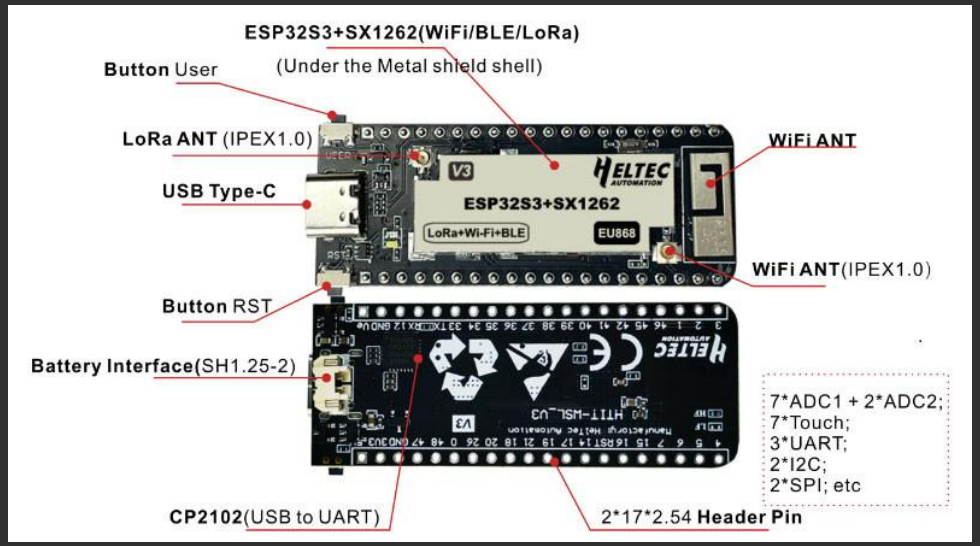


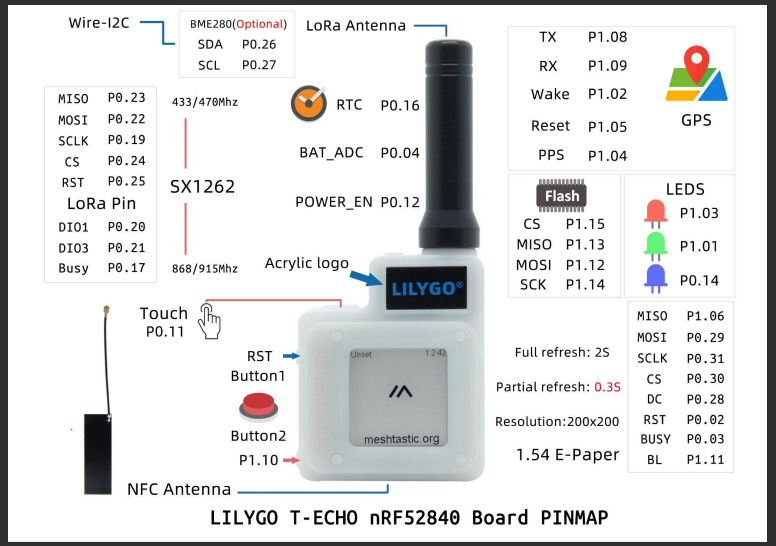


Defining Terms: Nodes

- **LoRa Node**: A combination of LoRa radio & host microcontroller. Together, they can run Meshtastic or Meshcore!

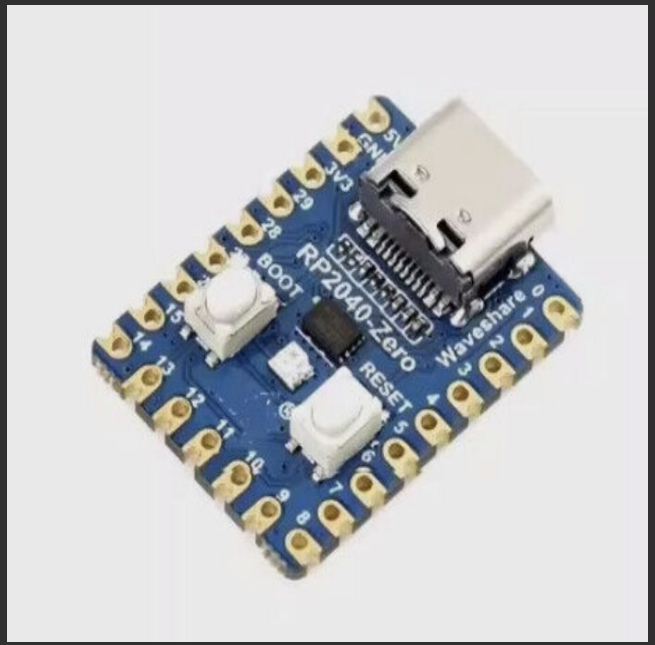
Simple Node <----------> Complex Node

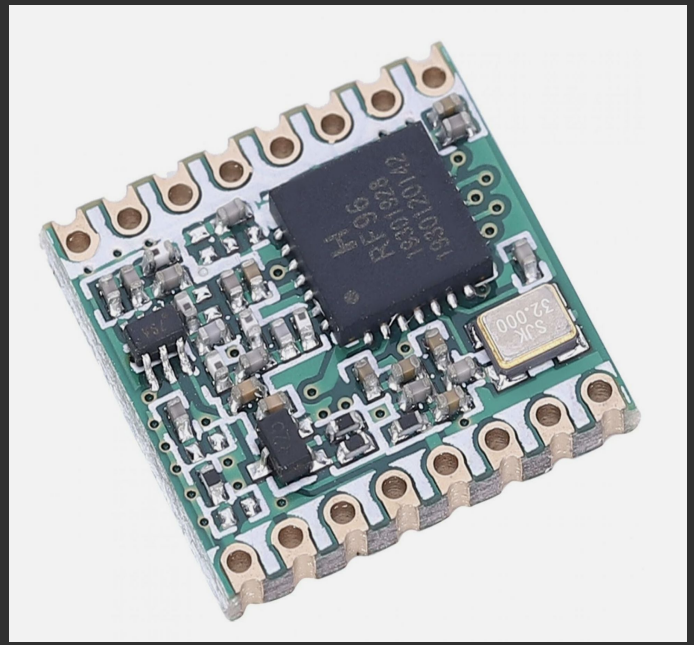


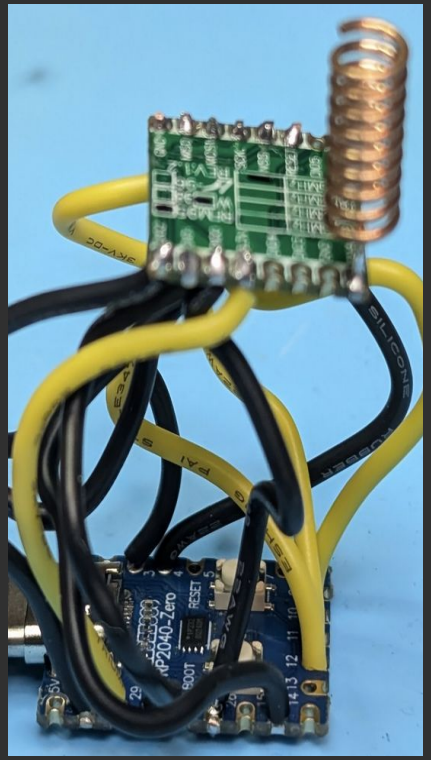


Host + Radio = Node

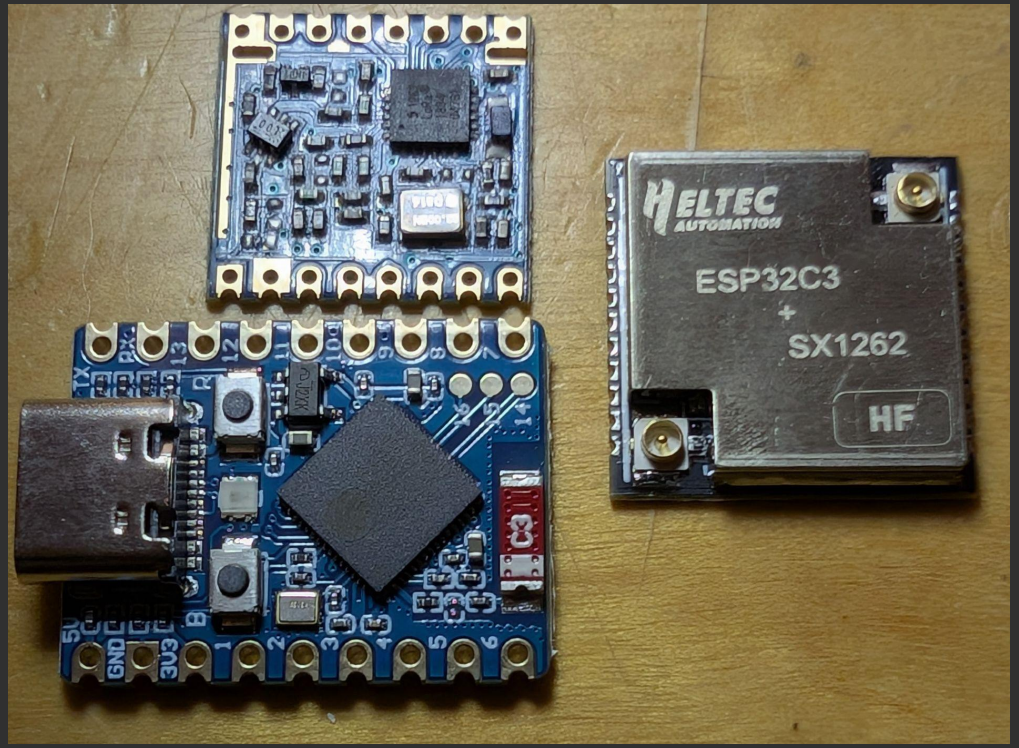
RP2040 Zero (host breakout) + RFM95 (radio breakout) = $5 Node



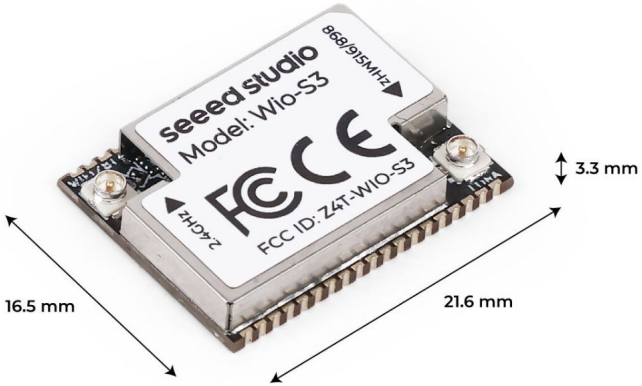




Future Nodes for Under $10



seeed studio



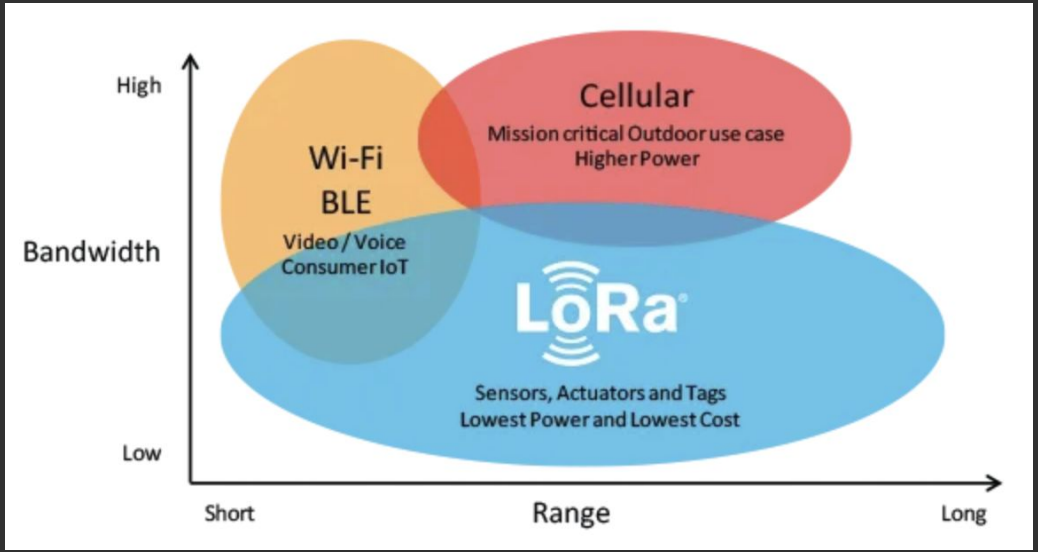
Wio-S3 Wireless Module with IPEX, ESP32-S3R8, Dual-core Xtensa® LX7 and SX1262 embedded, LoRaWAN, Wi-Fi & BLE, supports EU868 & US915

USD **$7.99**

Shipping & duties calculated at checkout.

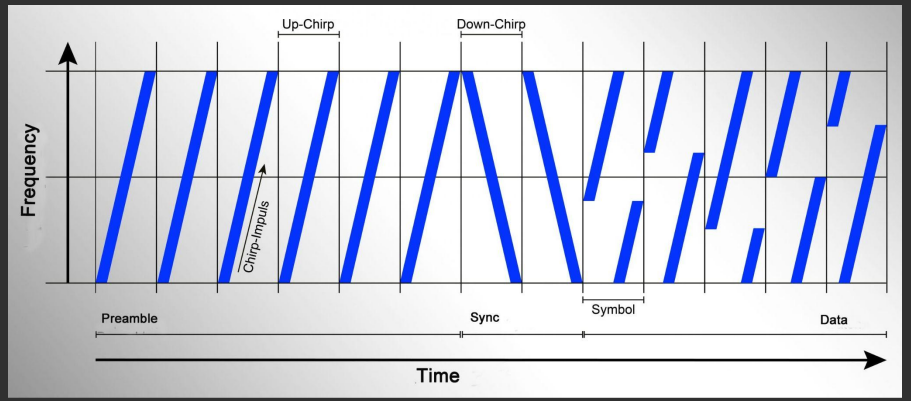
Introduction to LoRa

- LoRa stands for Long Range, alternative radio standard
- Radios that operate in unlicensed sub-GHz frequency bands
- Slow data rates but very long range
- Not for webcam monitoring, but ideal for telemetry, chat, control signals



What makes LoRa special?

- Long range protocol uses up and down “chirps” to send data
- Chirp Spread Spectrum modulation helps packets survive noise & interference
- Successfully receive & decode packets 20 dB below the noise level
- Receiver sensitivity of a maximum of -149 dBm



Data rate	Sensitivity CC1020 FSK	CC1101 FSK dBm	SX1261 FSK	SX1261 LoRa
1.2 kBit/s	-117 dBm	-112 dBm	-123 dBm	-129 dBm
2.4 kBit/s	-117 dBm	-111 dBm	-121 dBm	-126 dBm
4.8 kBit/s	-114 dBm	-109 dBm	-118 dBm	-123 dBm
9.6 kBit/s	-112 dBm	-107 dBm	-116 dBm	-120 dBm

Source: interline.pl

What other devices can I use?

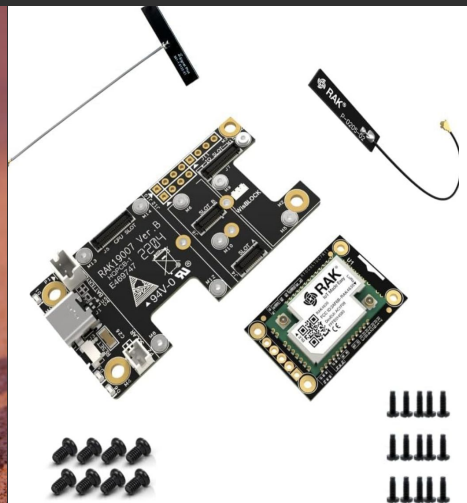
Here are some of the usual suspects:
Heltec v3, Lilygo T-Deck, Rak Wireless Wizblock,

Lilygo T-Deck, Nibble, Nugget

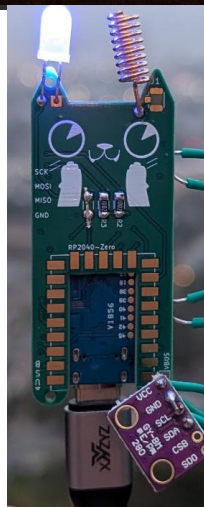












Regional frequencies - Check your radios!

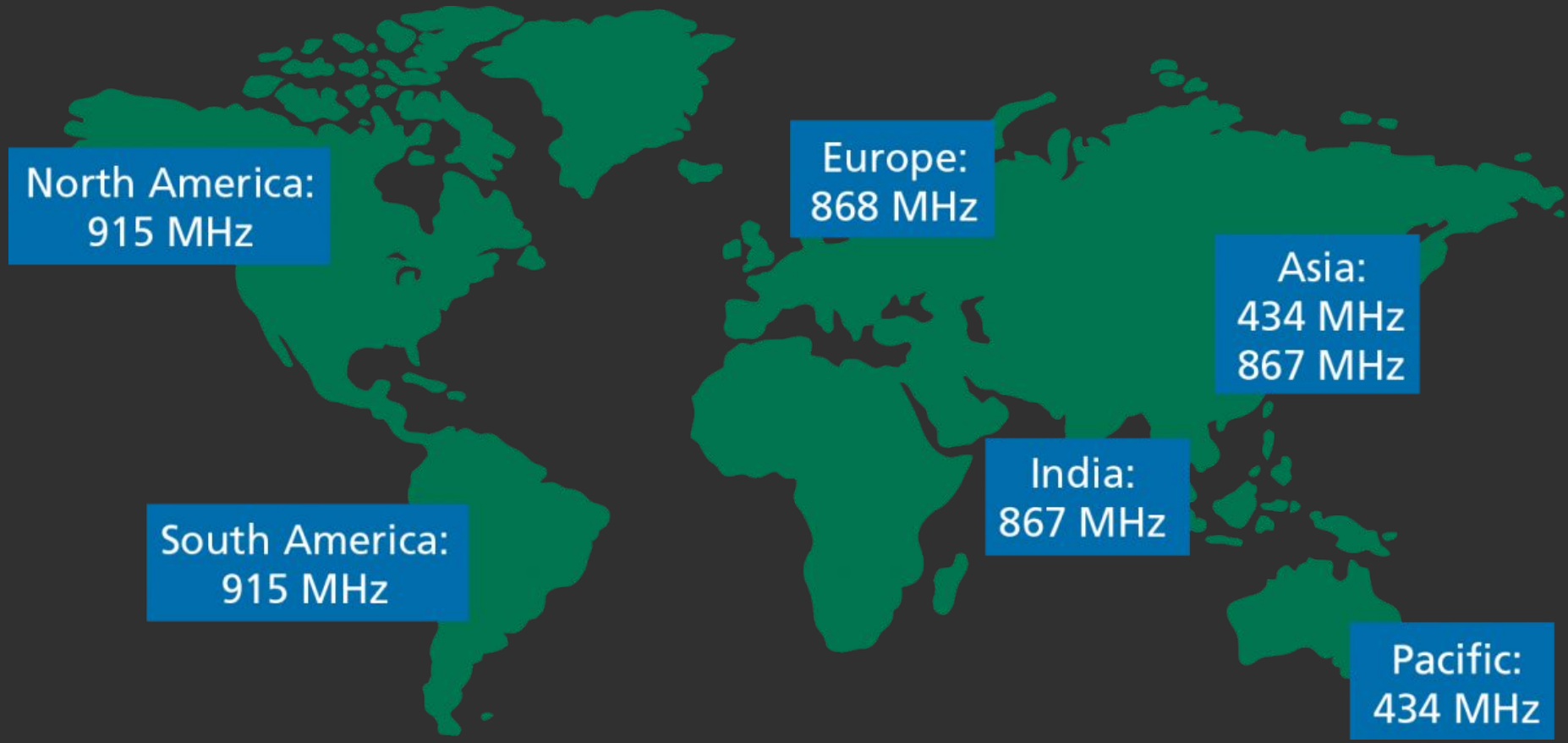
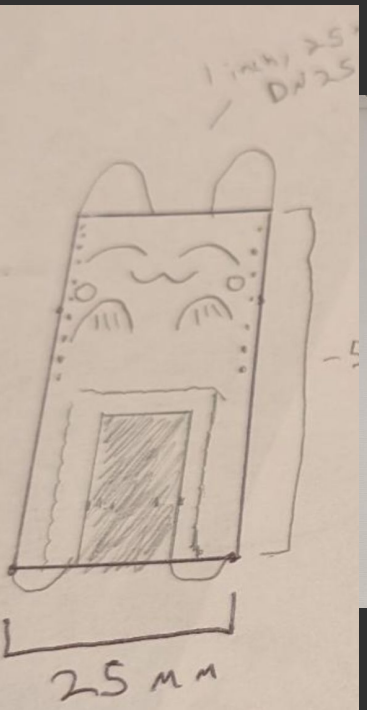
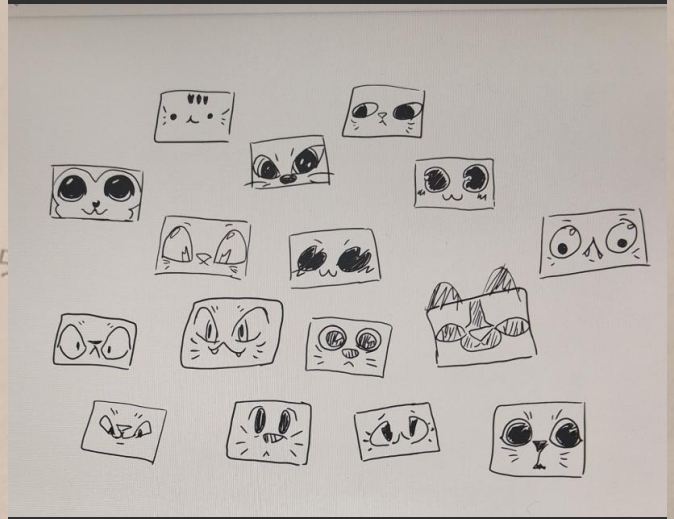
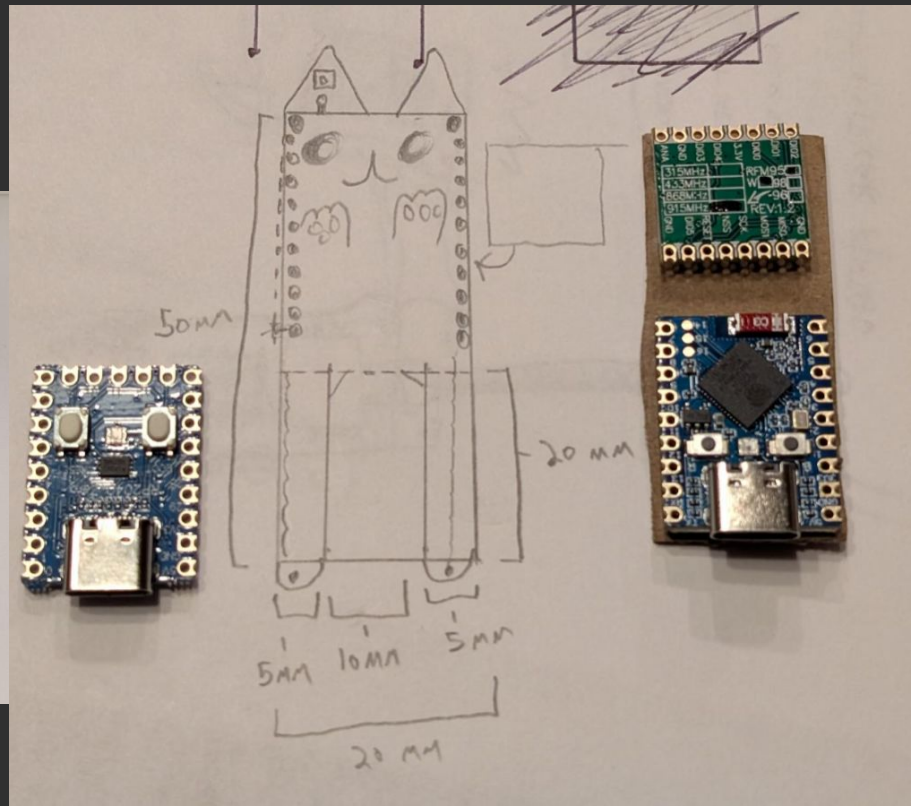


image from https://lowpowerlab.com

Making 300 Custom Meshtastic Kits: The Nibble



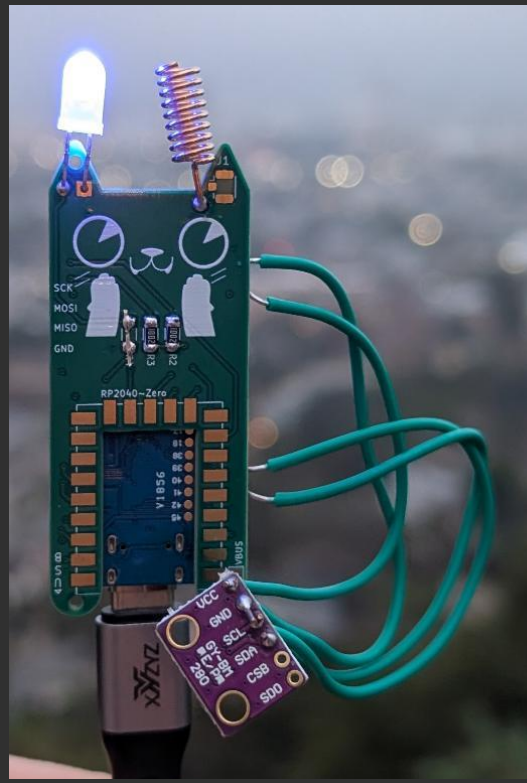




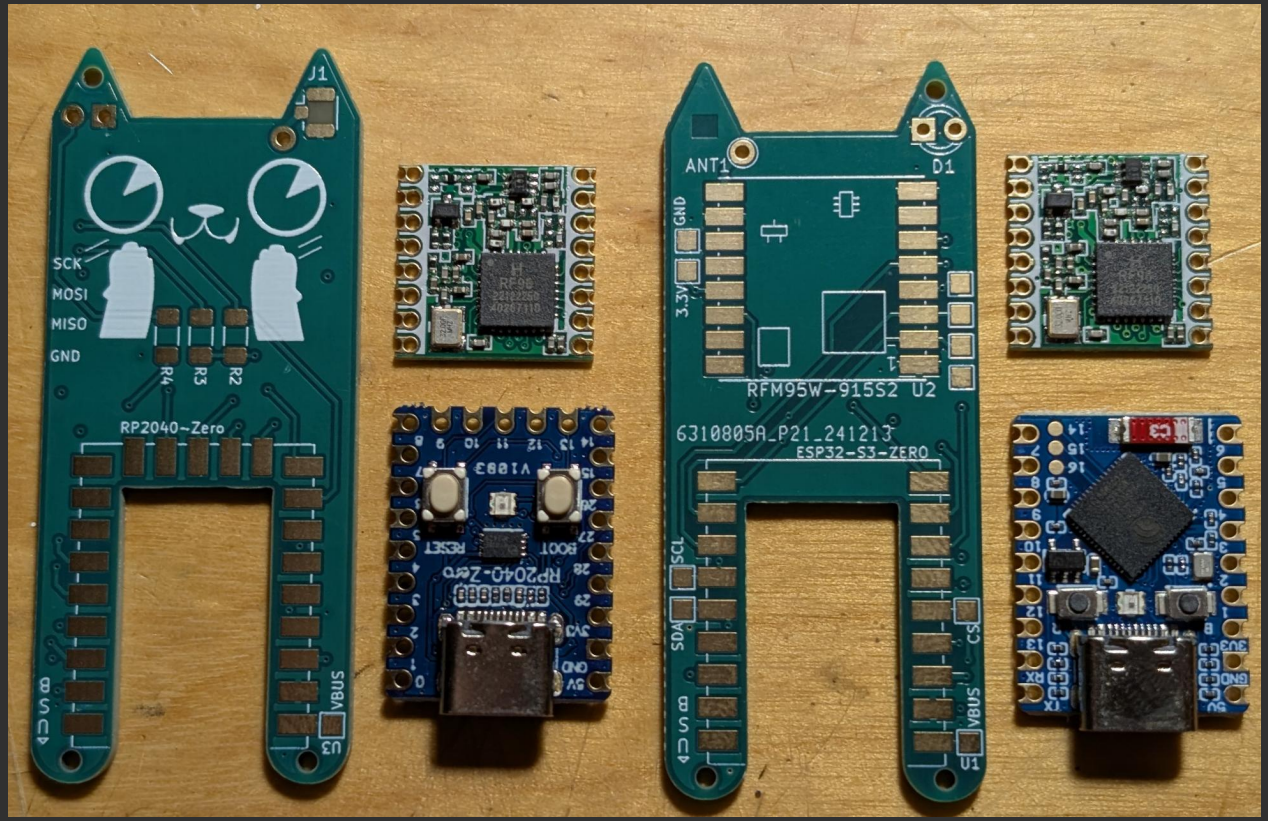
The Nibble - The Original Cat Meshtastic Node Kit

- Combines RP2040 or ESP32s3 host with sx1276 radio
- Designed to be soldered by beginners
- Select your micro: RP2040 with serial only, or esp32s3 with Bluetooth & Wi-Fi
- Designed to fit 1 inch PCV pipe for outdoor placement
- Pads for adding SPI or i2c sensor chains like bme280

- RP2040 version can’t connect to iOS over serial, but serial works on Android, desktop, & webapps



The Nibble - The Original Cat Meshtastic Node Kit



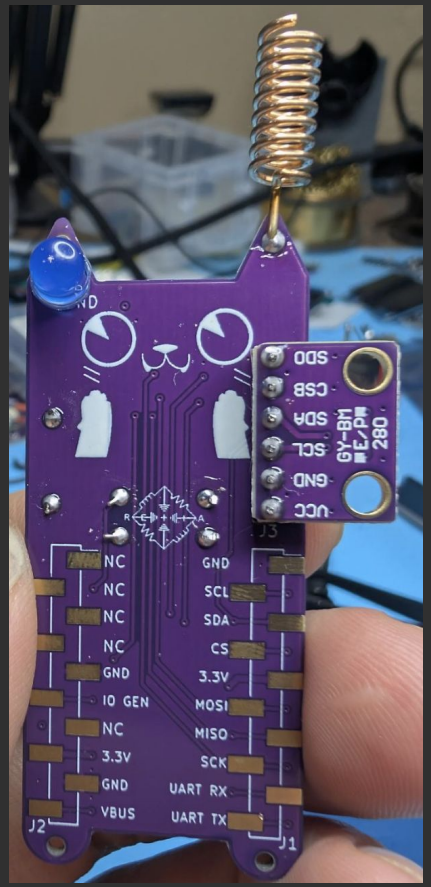
We open-sourced this design!

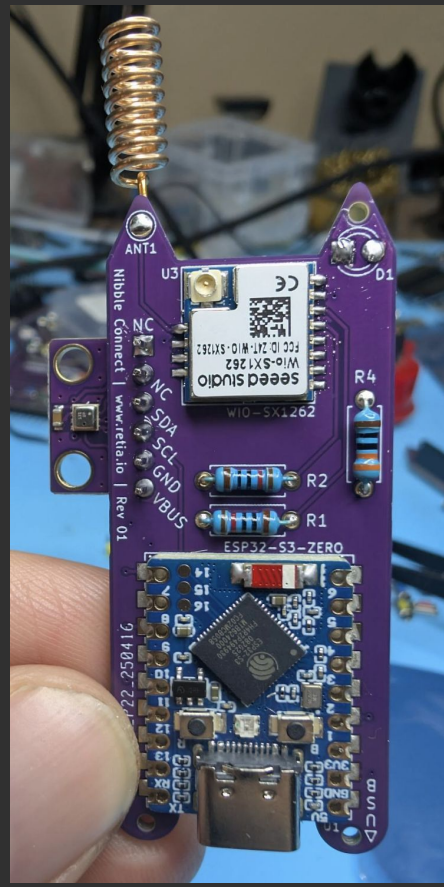
You can buy one from us, or you can make your own: github.com/retiallc/nibble

Update: The Nibble Connect

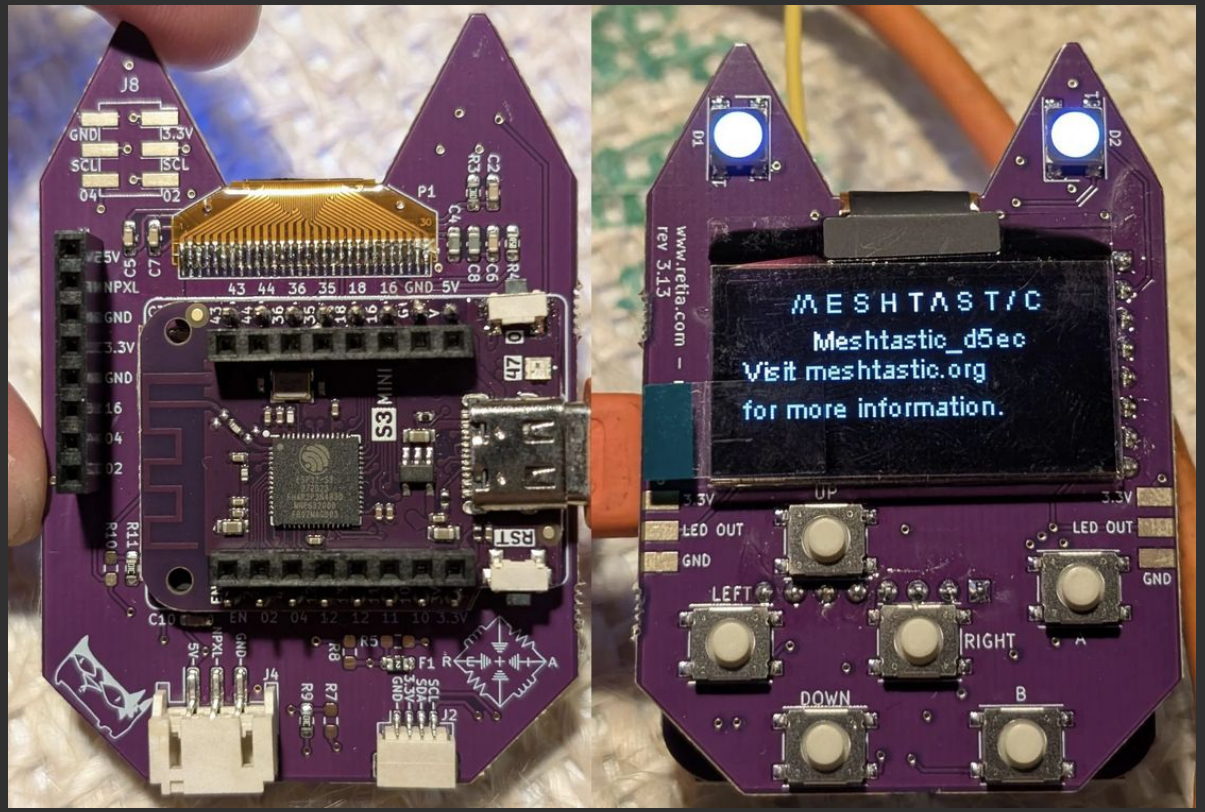
Based on feedback, we upgraded the Nibble for our kids cyber camps:

- Selected ESP32s3 Microcontroller (no more rp2040)
- Upgraded to Seeed sx1262 LoRa Radio - software definable frequency
- SMD Female Pin Headers
- Mount for BME280 weather sensor

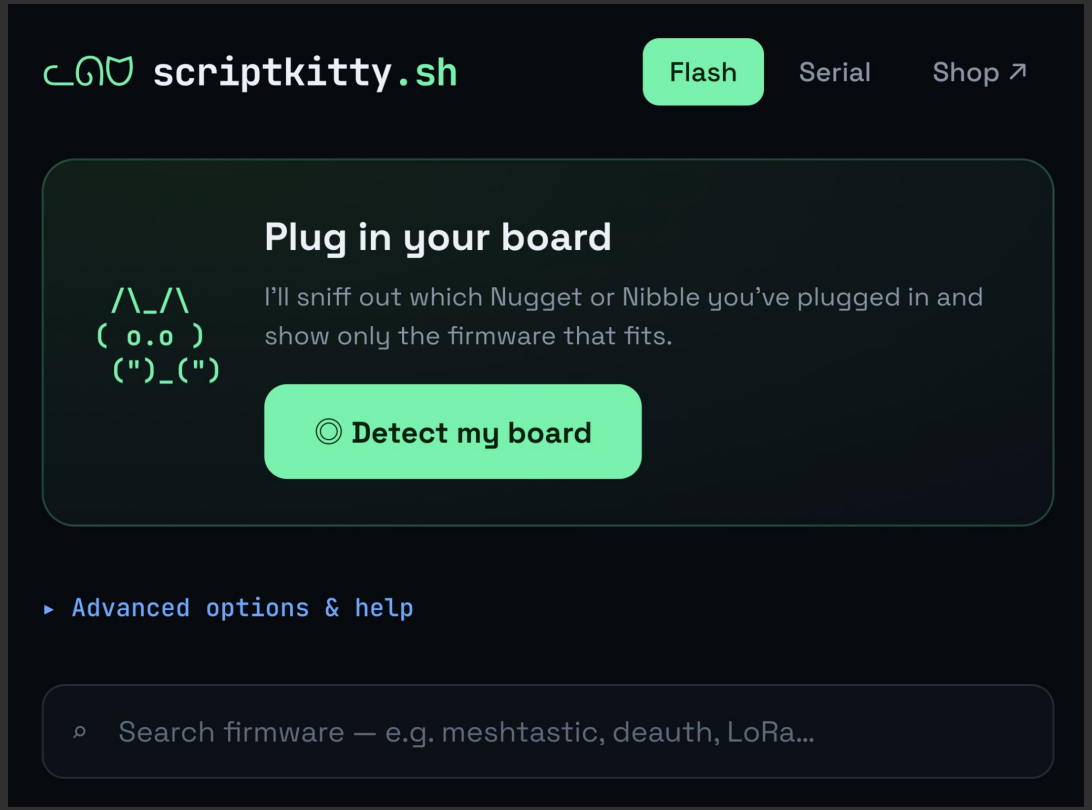




Bluetooth Nugget



Demo Time: Let’s Flash Meshtastic - www.scriptkitty.sh

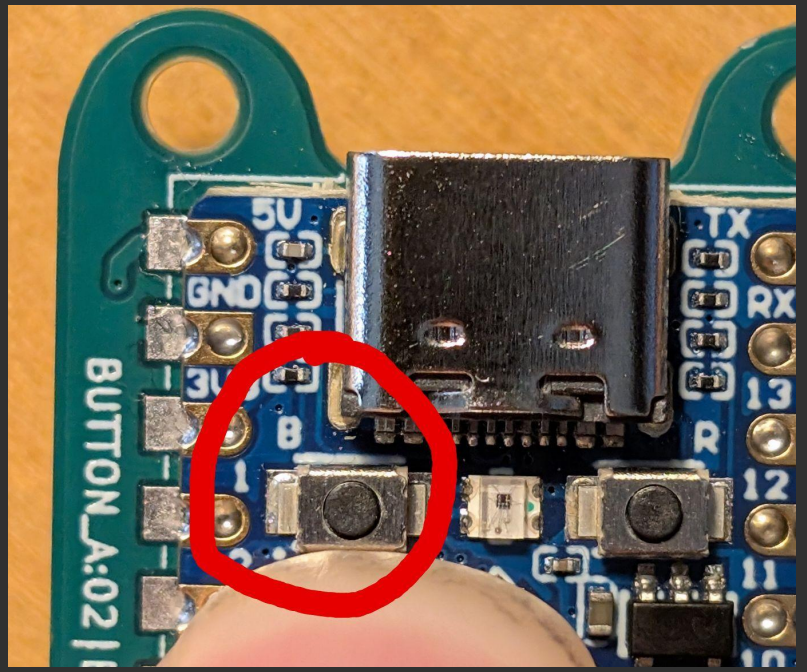


Flash Via Chrome-Based Browser

Hold down the “B” button while plugging your Nibble into your computer via USB type C

Once plugged in, click “Detect my board” and select your device.

If you’re not sure which it is, unplug and plug it back in to see it appear again.

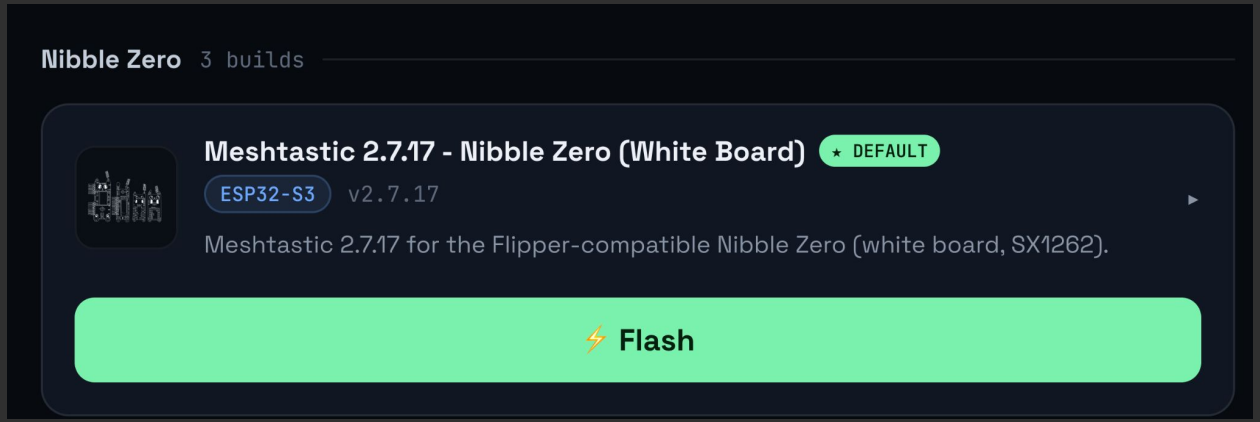


Flash Meshtastic

After you select your board, the scriptkitty.sh will auto-detect your board and show you projects you can flash.

Click “Flash” under “Meshtastic For Nibble Zero (White Board).

Once it’s done flashing, unplug it and plug it back in. That’s it!

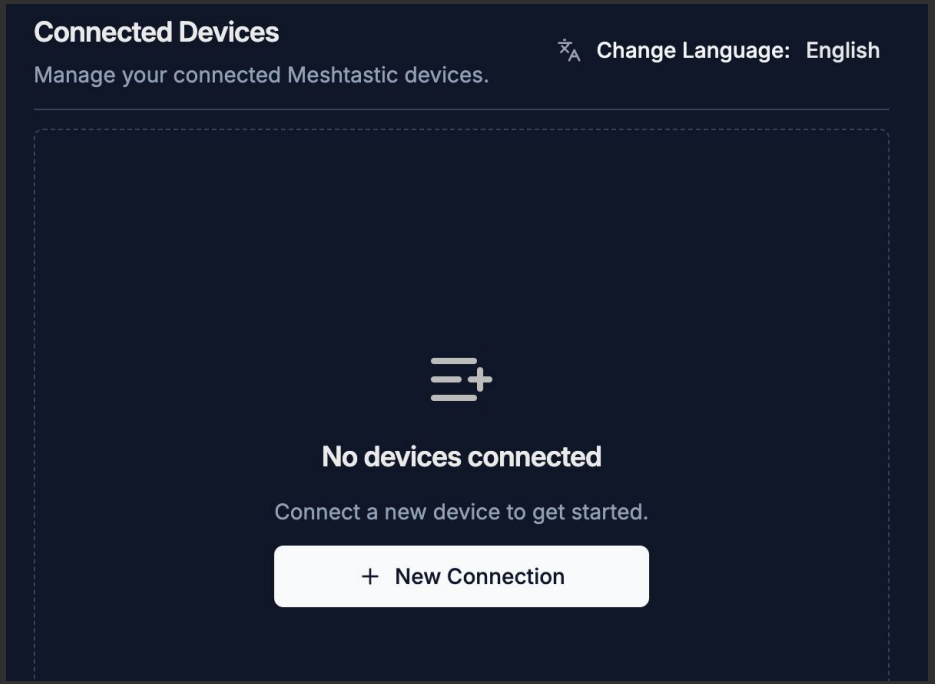


Check for Heartbeat

When you plug your Nibble back in, it should turn on the screen after 10 seconds

It will NOT transmit until you select the correct region, so let's do that!

In a browser, go to https://client.meshtastic.org/



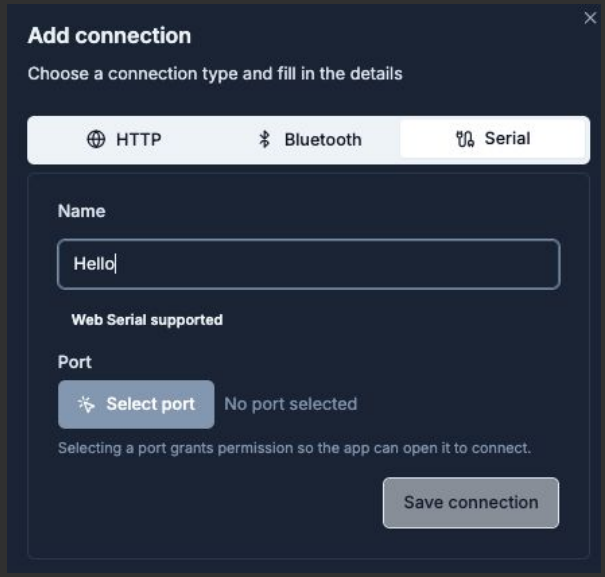
Connect Via Browser

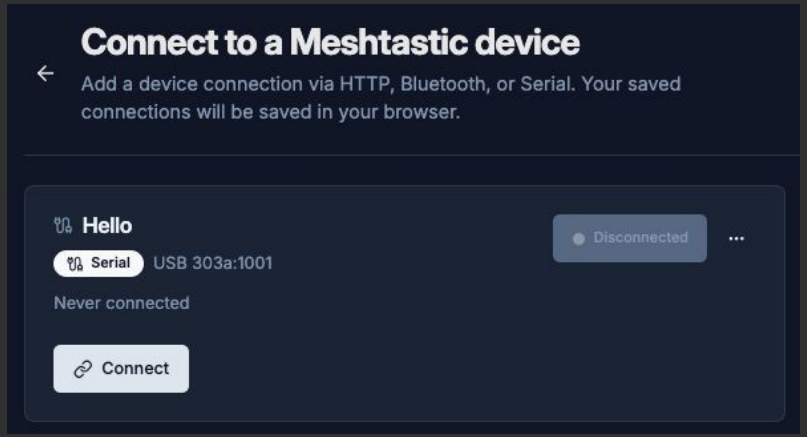
Click on “New Connection” and select “Serial” from the menu of options.

You can also try Bluetooth here if you want.

Click “New device” and select your Nibble the same way you did to flash it.

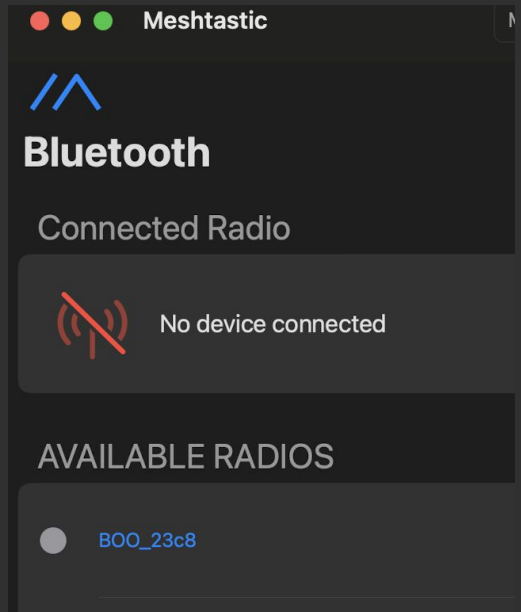
It should appear, click it to enter the main menu!

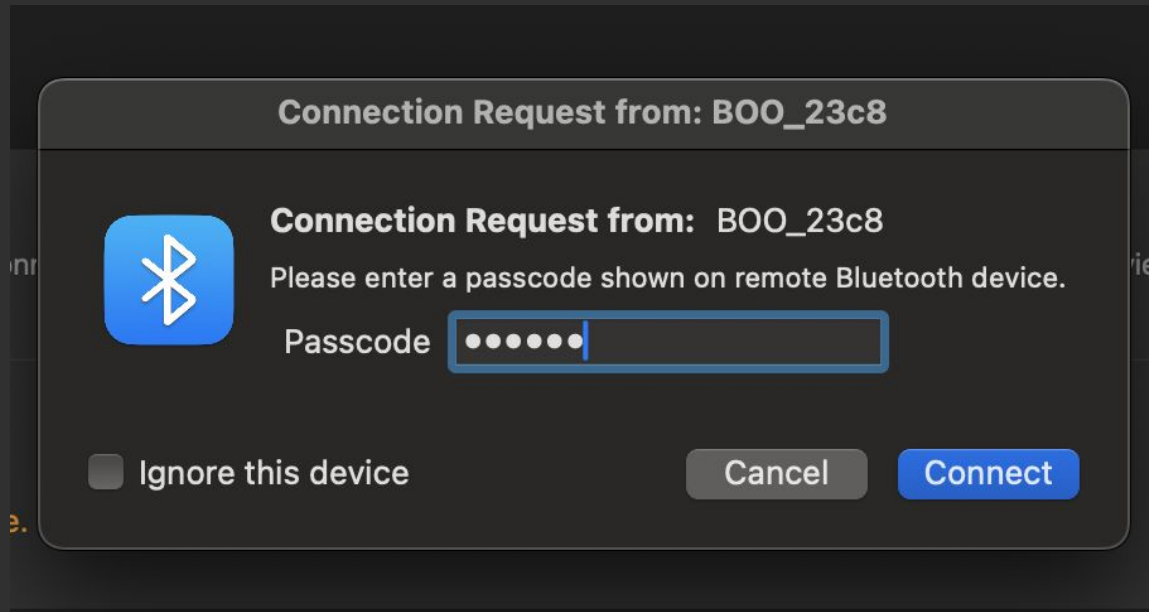




Alt: Connect via Bluetooth with Smartphone App

- Download the Meshtastic smartphone or Desktop app
- Look for your node short name via Bluetooth
- Pair using the pin that appears on the screen

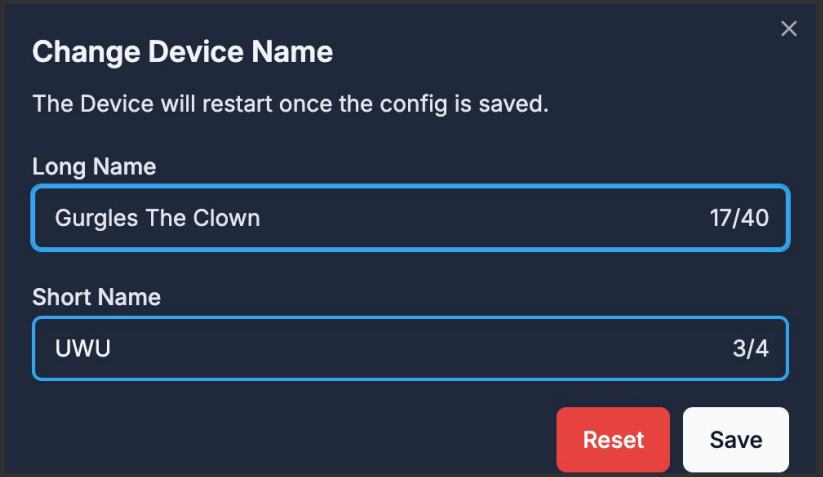


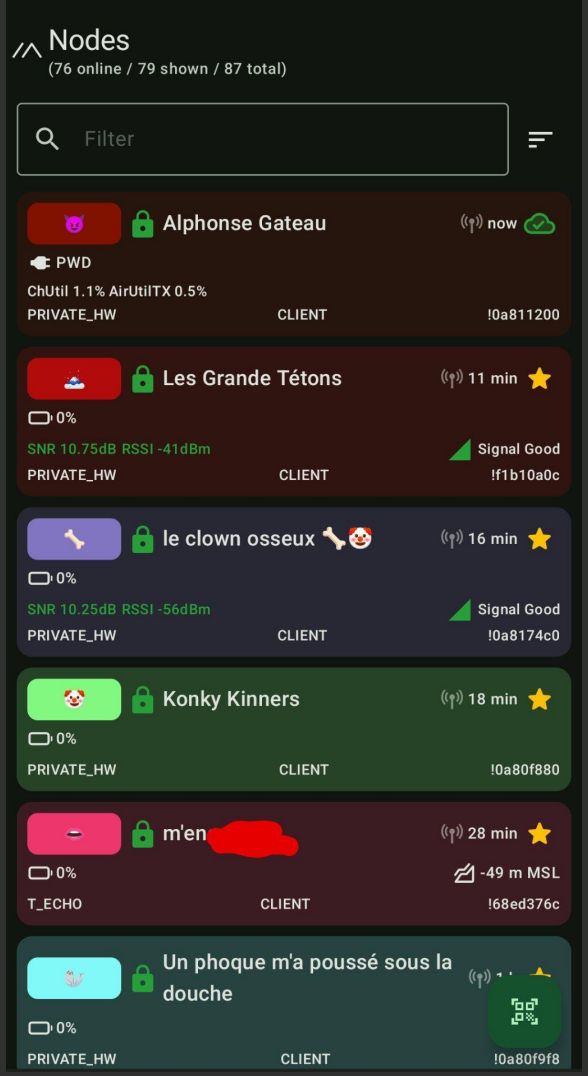


Change Device Name

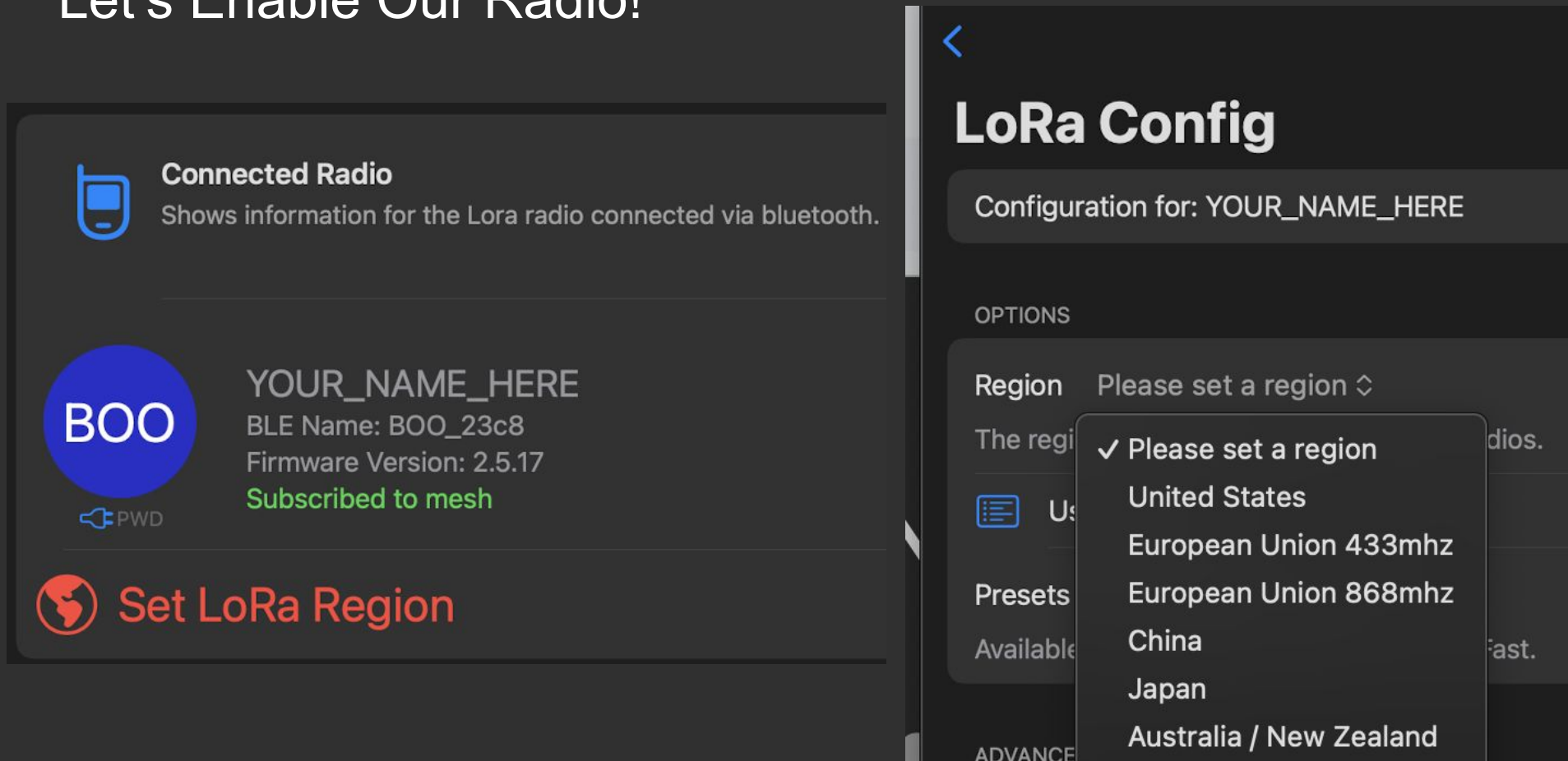
At the main menu, click “Change device name” and add a new name.
You can use emojis!

After you save, it will reboot with the new name.





Let’s Enable Our Radio!



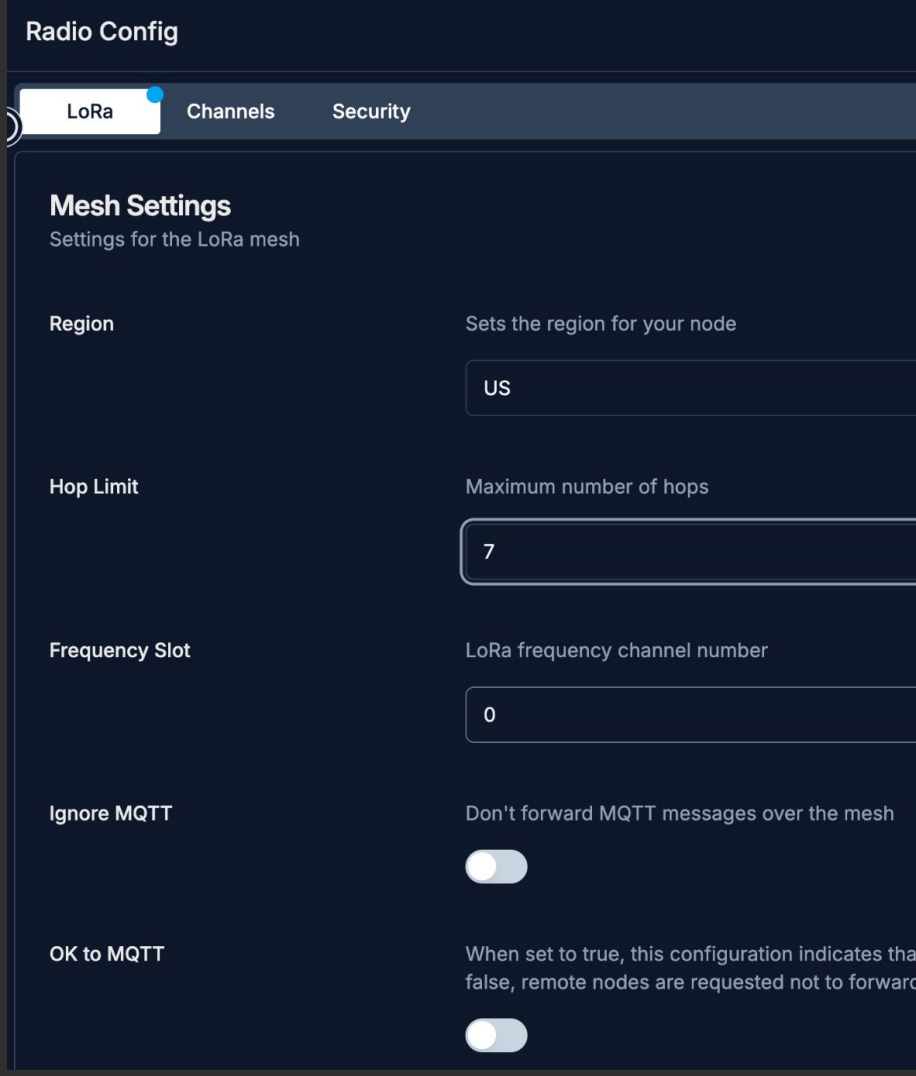
Set Region - Web or Buttons

You can use the screen & buttons to set the region, or:

Click on Config, then LoRa.

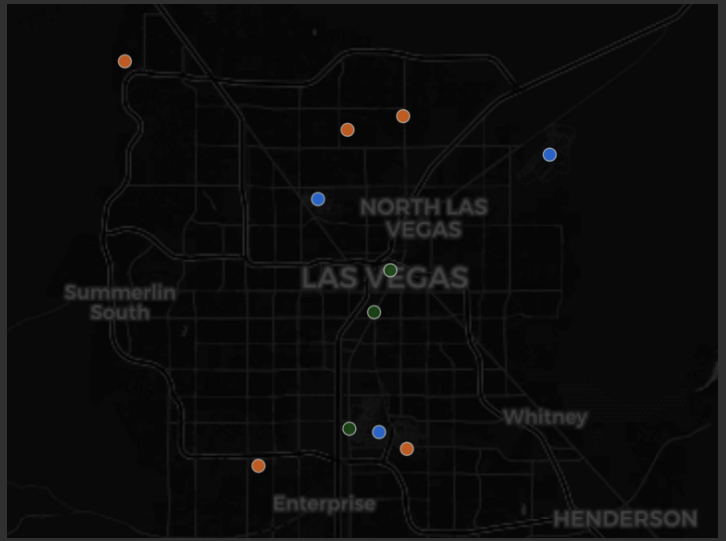
Set the correct region, and set the hop limit to 7.

Save, and your Nibble will reboot ready to transmit!



Las Vegas MeshNets

- **LASmesh - https://lasmesh.com**



Join LASmesh on Meshtastic

To connect with the community, scan the QR code below.

https://meshtastic.org/e/#Ci8SIGztXSO5VD8d2QrD_KIJyCkEopt-gop4Ql_uW3hZ-rE8GgdMQVNtZXNoOgIIIBIOCAE4AUAESAFQHlgUaAE

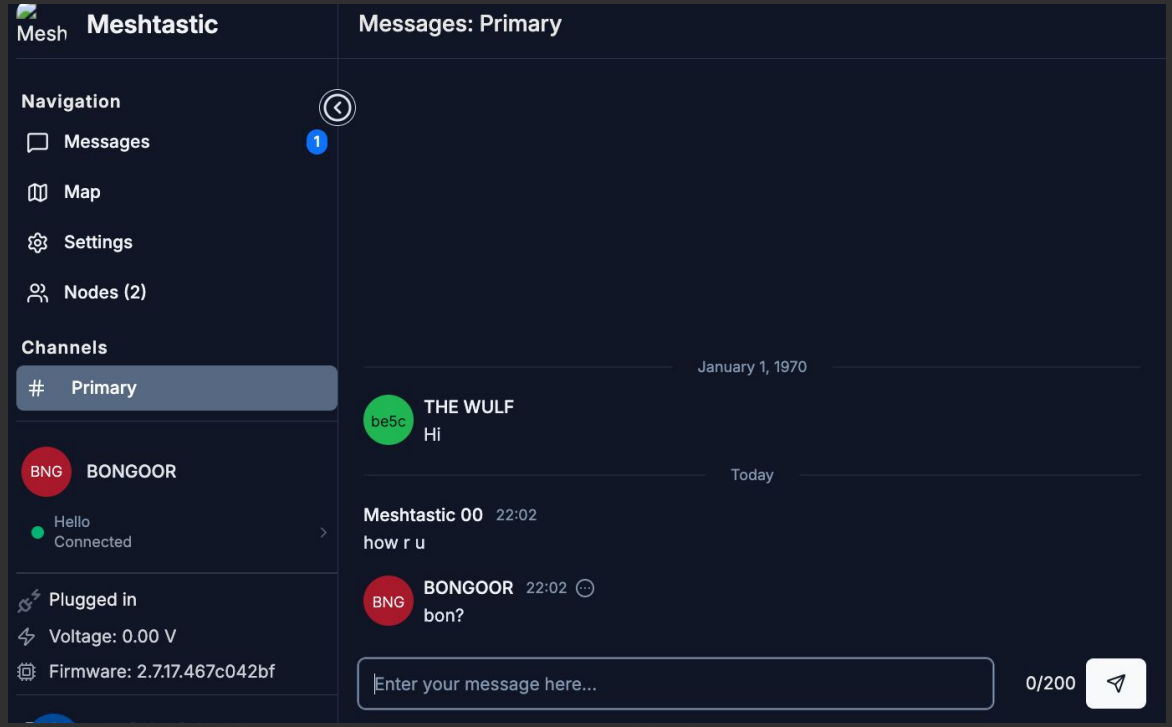
The **LASmesh channel** is a "sub chat" (outside of the public default channel). Feel free to say hello, send a `!ping` or `!test` connectivity here! The basic MeshBot is live! Type `!help` for more.

Be sure to select "**Add**" after scanning to apply the channel settings properly.

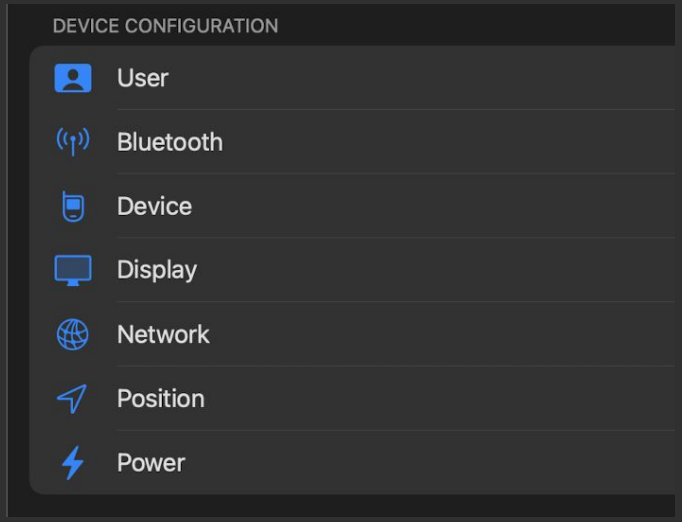
We are transmitting!

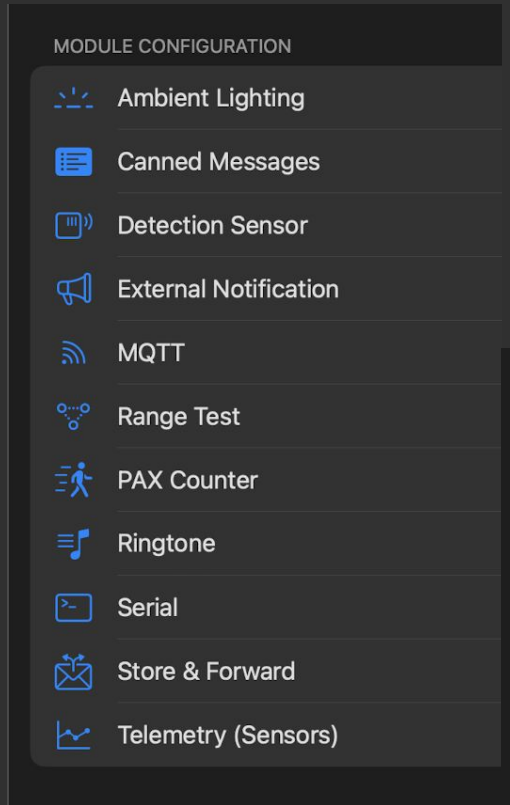
Go to channel, primary channel, and type a message.

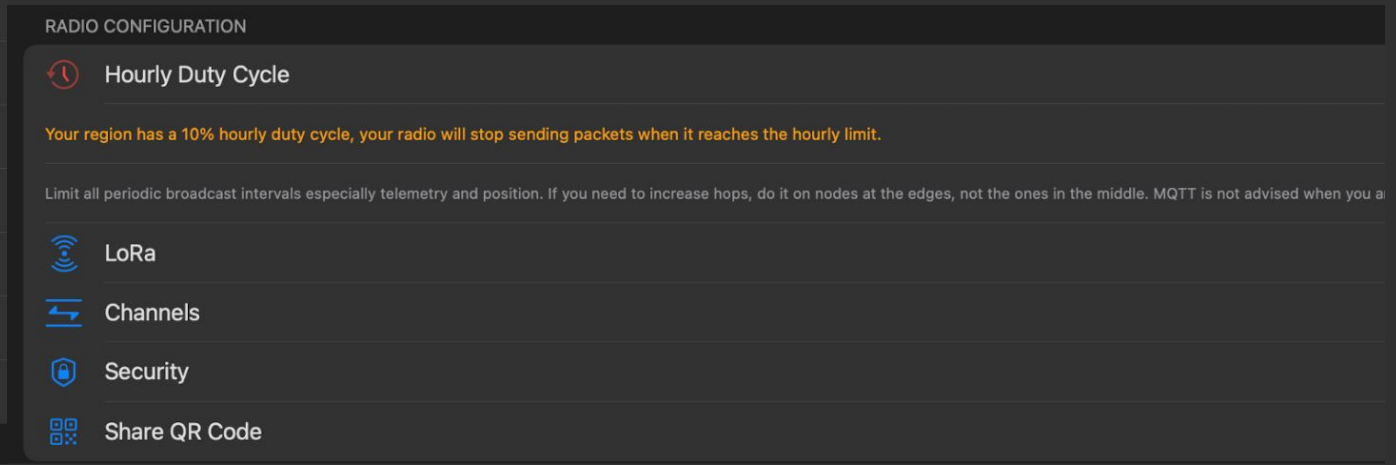
If you see “acknowledged,” another node has seen and retransmitted your message!



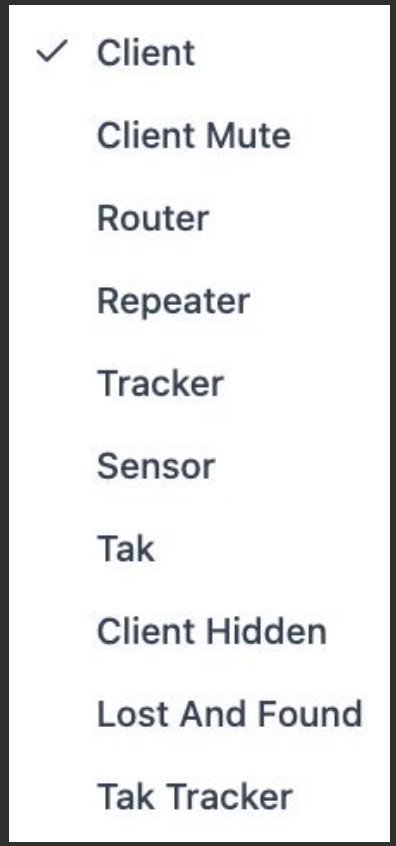
Let’s Explore Settings!







Device roles



Device Role	Description	Best Uses
CLIENT	App connected or stand alone messaging device. Rebroadcasts packets when no other node has done so.	General use for individuals needing to communicate over the Meshtastic network with support for client applications.
CLIENT_MUTE	Device that does not forward packets from other devices.	Situations where a device needs to participate in the network without assisting in packet routing, reducing network load.
CLIENT_HIDDEN	Device that only broadcasts as needed for stealth or power savings.	Use in stealth/hidden deployments or to reduce airtime/power consumption while still participating in the network.
CLIENT_BASE	Personal base station: always rebroadcasts packets from or to its favorited nodes. Handles all other packets like CLIENT.	Use for stronger attic/roof "base station" nodes to distribute messages more widely from your own weaker, indoor, or less-well-positioned nodes.
TRACKER	Broadcasts GPS position packets as priority.	Tracking the location of individuals or assets, especially in scenarios where timely and efficient location updates are critical.
LOST_AND_FOUND	Broadcasts location as message to default channel regularly to assist with device recovery.	Used for recovery efforts of a lost device.
SENSOR	Broadcasts telemetry packets as priority.	Deploying in scenarios where gathering environmental or other sensor data is crucial, with efficient power usage and frequent updates.
TAK	Optimized for ATAK system communication, reduces routine broadcasts.	Integration with ATAK systems (via the Meshtastic ATAK Plugin) for communication in tactical or coordinated operations.
TAK_TRACKER	Enables automatic TAK PLI broadcasts and reduces routine broadcasts.	Standalone PLI integration with ATAK systems for communication in tactical or coordinated operations.
REPEATER [1]	Infrastructure node for extending network coverage by always rebroadcasting packets once with minimal overhead. Not visible in Nodes list.	Best positioned in strategic locations to maximize the network's overall coverage. Device is not shown in topology.
ROUTER	Infrastructure node for extending network coverage by always rebroadcasting packets once. Visible in Nodes list.	Best positioned in strategic locations to maximize the network's overall coverage. Device is shown in topology.
ROUTER_LATE	Infrastructure node that always rebroadcasts packets once but only after all other modes, ensuring additional coverage for local clusters. Visible in Nodes list. More details in our blog: Demystifying Router Late	Ideal for covering dead spots or ensuring reliability for a cluster of nodes where placement doesn't benefit the broader mesh. Device is shown in topology.

What device roles mean

Device Role	BLE/WiFi/Serial	Screen Enabled	Power Consumption	Retransmit	Prioritized Routing	Visible in Nodes List
CLIENT	Yes	Yes	Regular	Yes	No	Yes
CLIENT_MUTE	Yes	Yes	Lowest	No	No	Yes
CLIENT_HIDDEN	Yes	Yes	Lowest	Local Only	No	No
CLIENT_BASE	Yes	Yes	Regular	Yes	Favorites Only[2]	Yes
TRACKER	Yes	No	Regular / Low	Awake Only[3]	No	Yes
LOST_AND_FOUND	Yes	No	Regular	Yes	No	Yes
SENSOR	Yes	No	Regular / Low	Awake Only[3]	No	Yes
TAK	Yes	Optional	Regular	Yes	No	Yes
TAK_TRACKER	Yes	Optional	Regular	Yes	No	Yes
ROUTER	No[4]	No	High	Yes	Yes	Yes
ROUTER_LATE	Yes	Yes	High	Yes	No	Yes
REPEATER	Yes	No	High	Yes	Yes	No

Surprise: changing to a router will turn off BLE!

Meshtastic LoRa Radio Profiles

8 Profiles: condense complex settings into common use cases

Trade-offs between long range or short range benefits (speed/range)

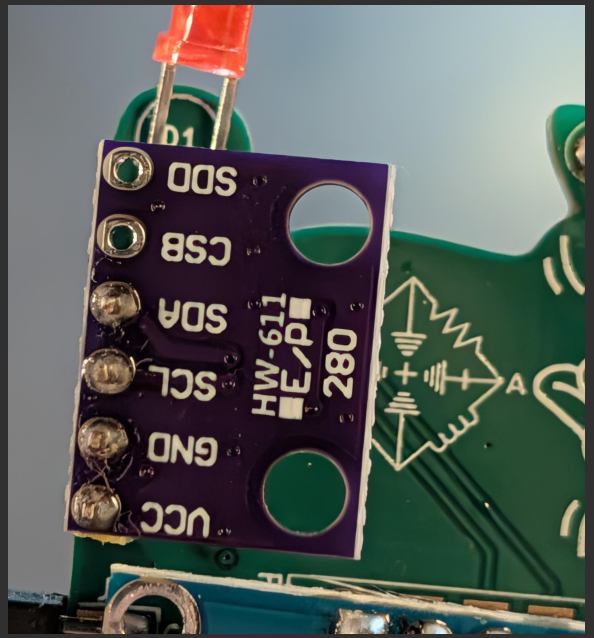
Often customized for conferences and events

Channel setting	Alt Channel Name	Data-Rate	SF / Symbols	Coding Rate	Bandwidth	Link Budget
Short Range / Turbo	Short Turbo	21.88 kbps	7 / 128	4/5	500 kHz[1]	140dB
Short Range / Fast	Short Fast	10.94 kbps	7 / 128	4/5	250 kHz	143dB
Short Range / Slow	Short Slow	6.25 kbps	8 / 256	4/5	250 kHz	145.5dB
Medium Range / Fast	Medium Fast	3.52 kbps	9 / 512	4/5	250 kHz	148dB
Medium Range / Slow	Medium Slow	1.95 kbps	10 / 1024	4/5	250 kHz	150.5dB
Long Range / Fast	Long Fast	1.07 kbps	11 / 2048	4/5	250 kHz	153dB
Long Range / Moderate	Long Moderate	0.34 kbps	11 / 2048	4/8	125 kHz	156dB
Long Range / Slow	Long Slow	0.18 kbps	12 / 4096	4/8	125 kHz	158.5dB
Very Long Range / Slow	Very Long Slow	0.09 kbps	12 / 4096	4/8	62.5 kHz	161.5dB

Adding sensors & hardware

Temperature,motion, other sensors are easy!

Enable Telemetry module & add common i2c sensors!



Telemetry Settings
Settings for the Telemetry module

Query Interval
Interval to get telemetry data
0 Seconds

Update Interval
How often to send Metrics over the mesh
0 Seconds

Module Enabled
Enable the Environment Telemetry

I2C Sensors Supported

Currently Supported Sensor Types

Sensor	I²C Address	Data Points
BMP085	0x76, 0x77	Temperature and barometric pressure
BMP180	0x76, 0x77	Temperature and barometric pressure
BMP280	0x76, 0x77	Temperature and barometric pressure
BME280	0x76, 0x77	Temperature, barometric pressure and humidity
BME68x	0x76, 0x77	Temperature, barometric pressure, humidity and air resistance
MCP9808	0x18	Temperature
INA260	0x40, 0x41, 0x43	Current and Voltage
INA219	0x40, 0x41, 0x43	Current and Voltage
INA3221	0x42	3-channel Current and Voltage
LPS22	0x5D, 0x5C	Barometric pressure
SHTC3	0x70	Temperature and humidity
SHT31	0x44	Temperature and humidity
PMSA003I	0x12	Concentration units by size and particle counts by size
DFROBOT_LARK	0x42	Temperature, barometric pressure, humidity, wind direction, wind speed
MAX30102	0x57	Heart Rate, Oxygen Saturation, and body temperature
MLX90614	0x5A	Body temperature

Modules (Part 1)

Name	Description
Ambient Lighting	Adjust the brightness of NCP5623 I2C RGB LEDs
Audio	Enable Support for Codec2 Voice Comms on certain devices.
Canned Message	Set a number of predefined messages to send out directly from the device with the use of an input device like a rotary encoder.
Detection Sensor	Configure a GPIO pin to be monitored for specified high/low status and send text alerts.
External Notification	Incoming messages are able to alert you using circuits you attach to the device (LEDs, Buzzers, etc).
MQTT	Forward packets along to an MQTT server. This allows users on the local mesh to communicate with users on another mesh over the internet.
Neighbor Info	Send info on 0-hop neighbors to the mesh.
Paxcounter	Count the number of BLE and Wifi devices passing by a node.

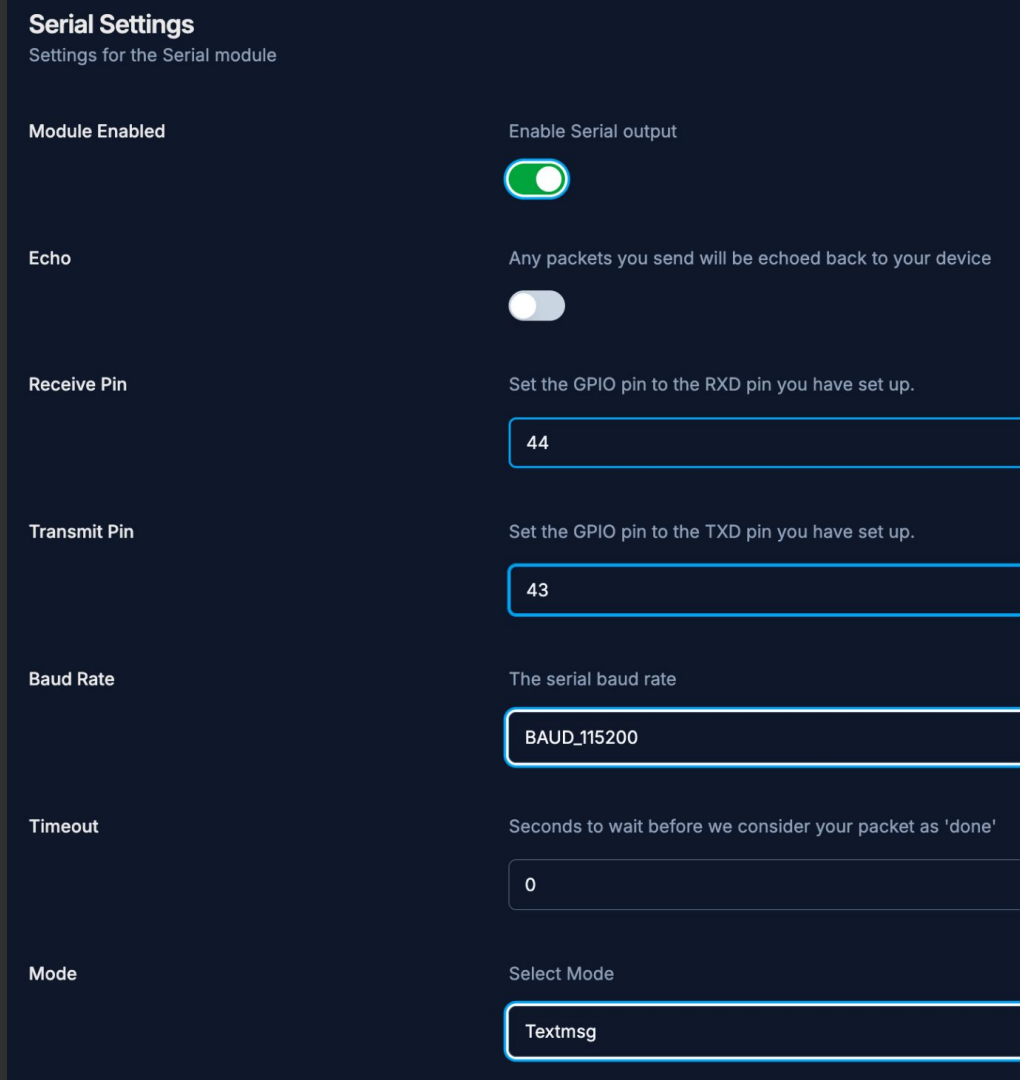
Modules (Part 2)

Range Test	Send messages with GPS location at an interval to test the distance your devices can communicate. Requires (at least) one device set up as a sender and one as a receiver. The receiver(s) will log all incoming messages to a CSV.
Remote Hardware	Set and read a GPIO status remotely over the mesh.
Serial Module	Send messages across the mesh by sending strings over a serial port.
Store & Forward	Stores messages on a device for delivery after disconnected clients rejoin the mesh.
Telemetry	Attach sensors to the device and transmit readings on a regular interval to the mesh.
Traceroute	Track which nodes are used to hop a message to a certain destination.

Enable Serial Module

Open the Nibble's Serial Module in Meshtastic and apply these settings:

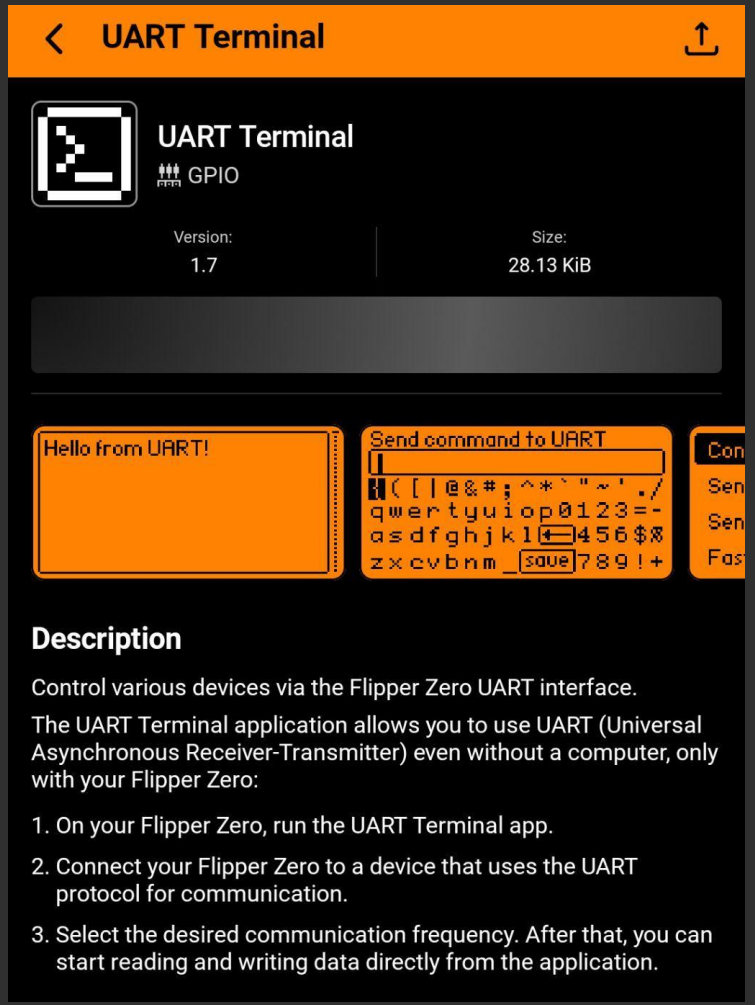
- Enable Serial Module
- Set Receive Pin to 44
- Set Transmit Pin to 43
- Set Baud Rate to 115200
- Set "Mode" to "Textmsg"



Let’s Set Up For Our Flipper!

On your Flipper app, Download the UART Terminal app.

It’s usually included in Momentum and Unleashed firmware



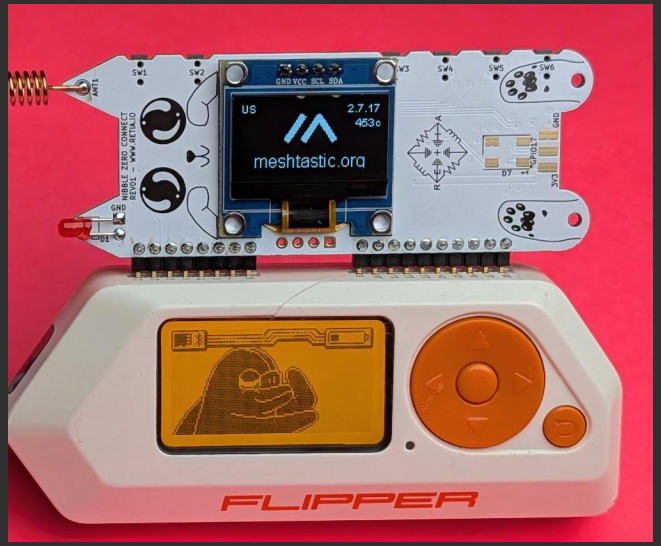
Plug In And Power On

Plug your Nibble into your Flipper.

To power your Nibble:

Either plug in your Flipper to power, or go to the GPIO app and set “5V on GPIO” to “on”

If it won’t load, make sure you flashed the most recent firmware.





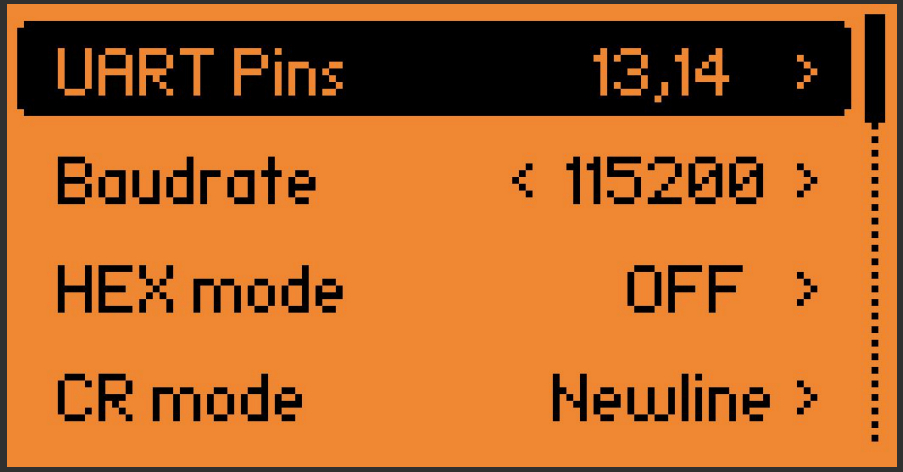
Open UART Terminal

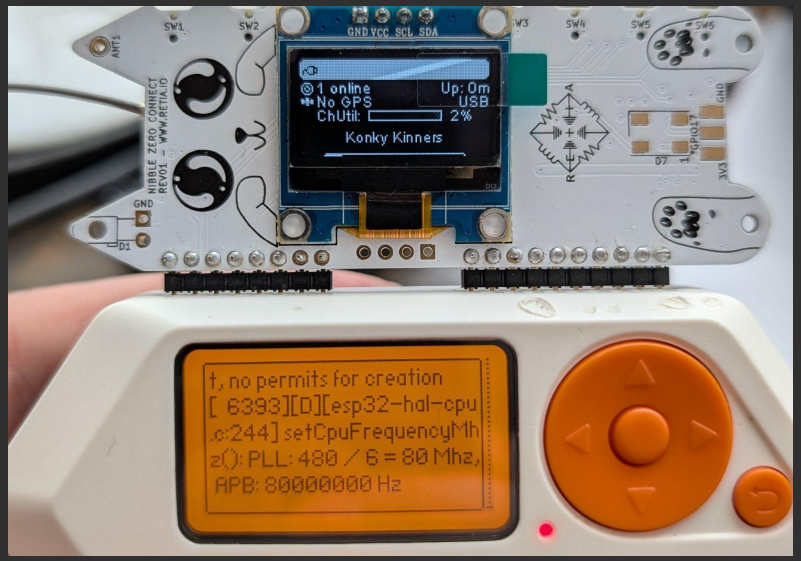
Under Setup, make sure the baud rate is set to 115200

Select “Open Port” and hit the reset button (R) on your Nibble.

You should see serial output appear on the screen!

This means it’s set up correctly.



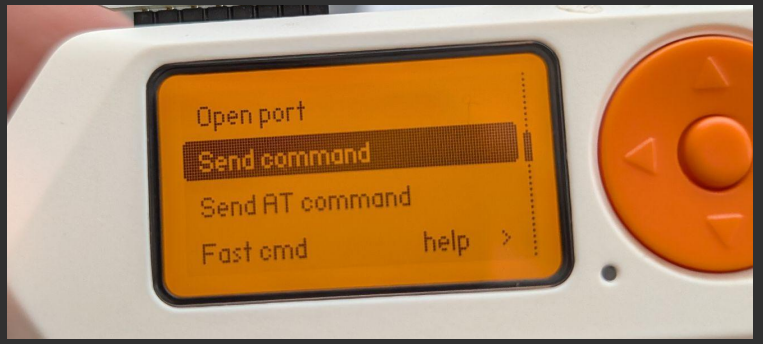


Send A Message

Select “Send Command”

Type out your message and hit “save”

Your message is sent! You should see it on other Meshtastic devices.

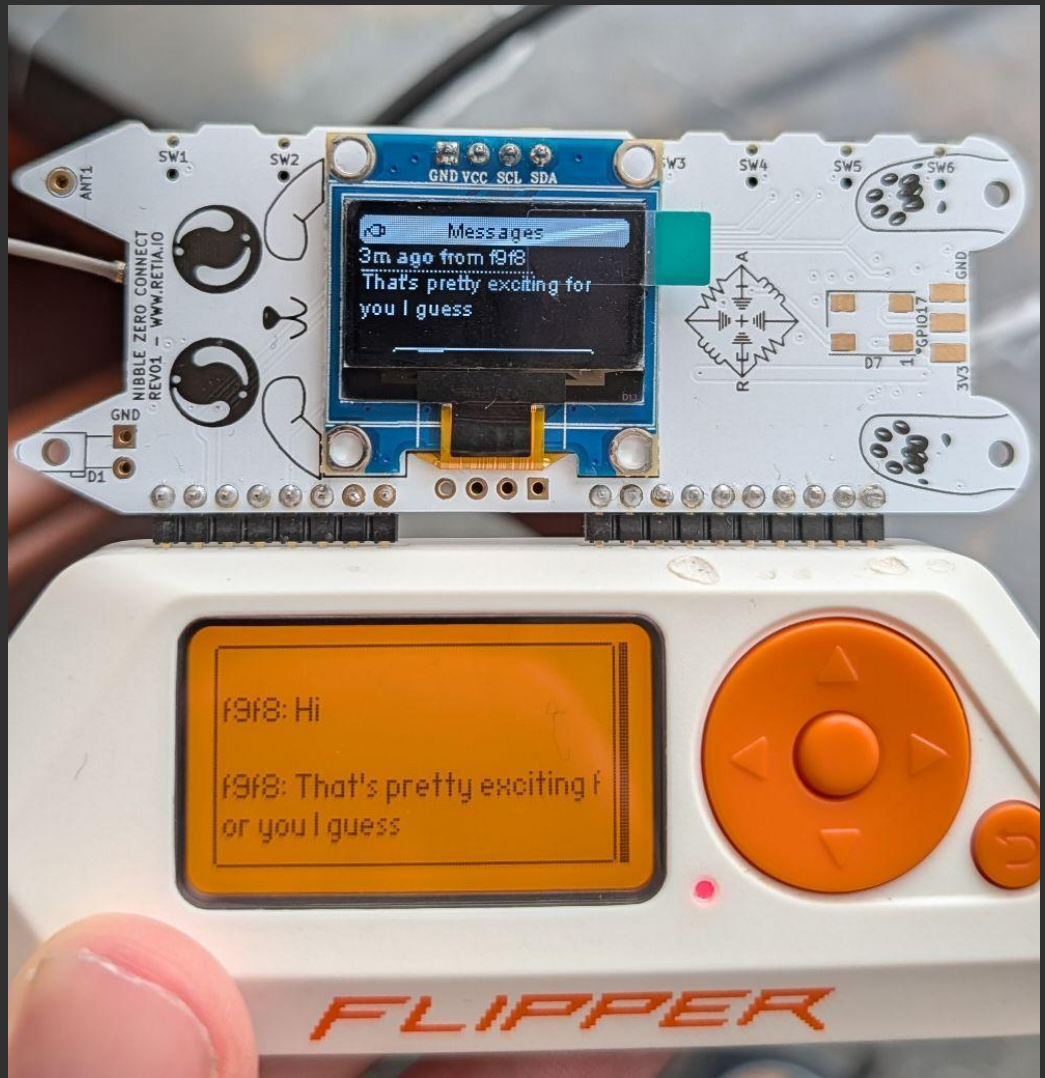






See Replies

Replies you receive will appear on the screen of your Flipper!



Want to go further?

https://github.com/cool4uma/UART_Terminal

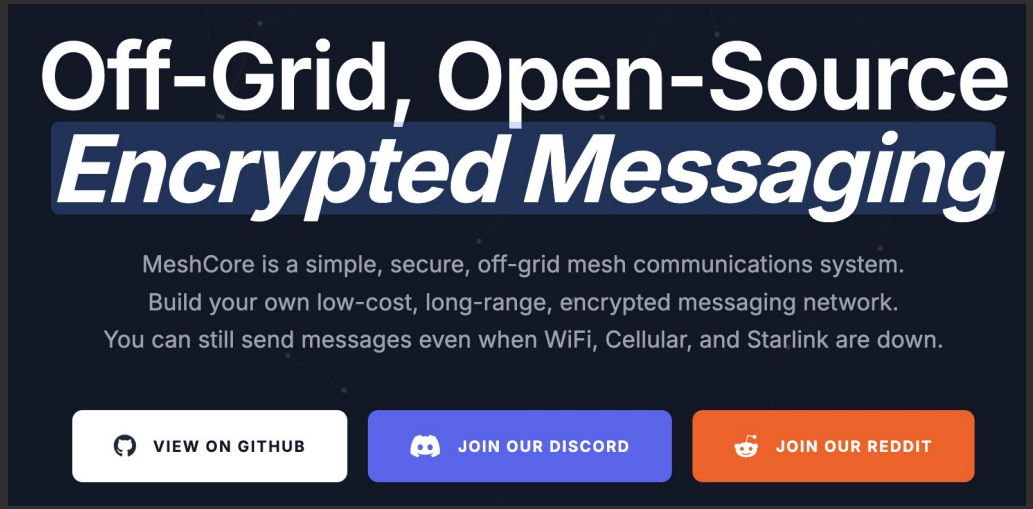
Remote control app is open source, you can improve and refine it for whatever use case you want!

Let’s Set Up Meshcore

In San Diego, Meshcore is finding more and more use!

Let’s set up Meshcore and join!

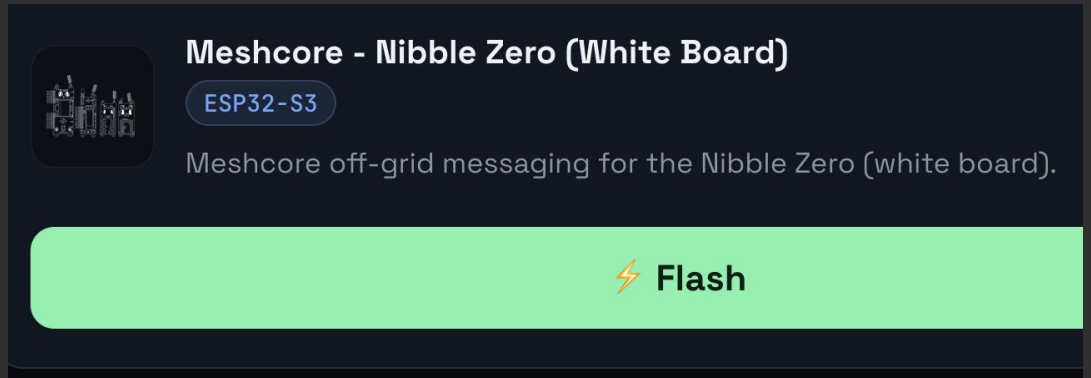
Download the MeshCore app for your smartphone



Flash on Scriptkitty.sh

Flashing is the same process as before on Scriptkitty.sh

1) Hold down the “B” button of your Nibble while plugging in
2) On Nugget.dev from a chrome browser, click “Connect your Nugget” and select your device from the drop-down menu
3) Select “Meshcore - **Nibble Zero**” from the firmware list
4) Click “Program” and unplug when your device is done flashing.

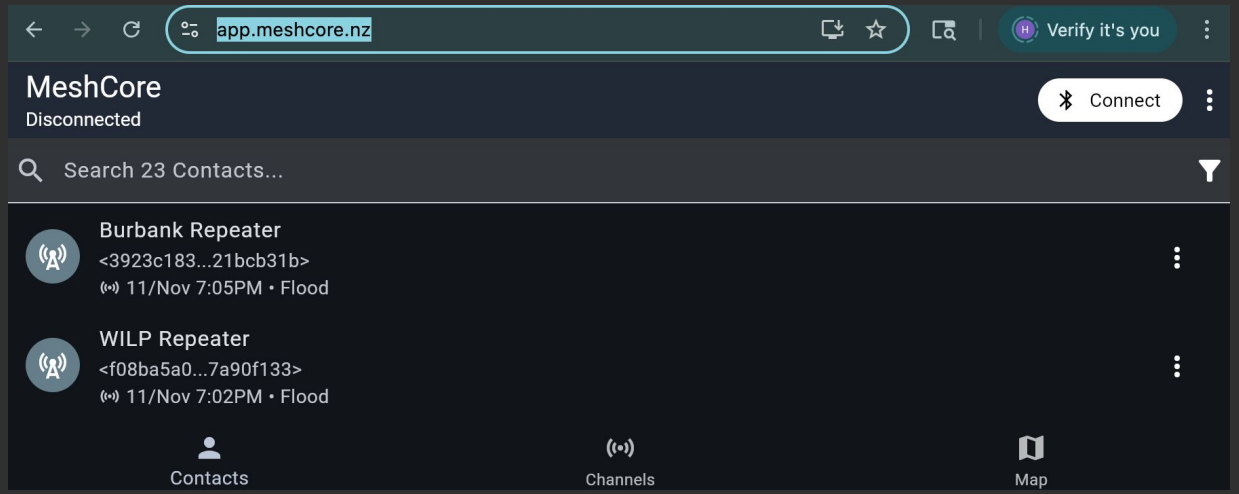


Plug In Your Device & Connect

Plug in your flashed device

It will start up and show a bluetooth pairing PIN

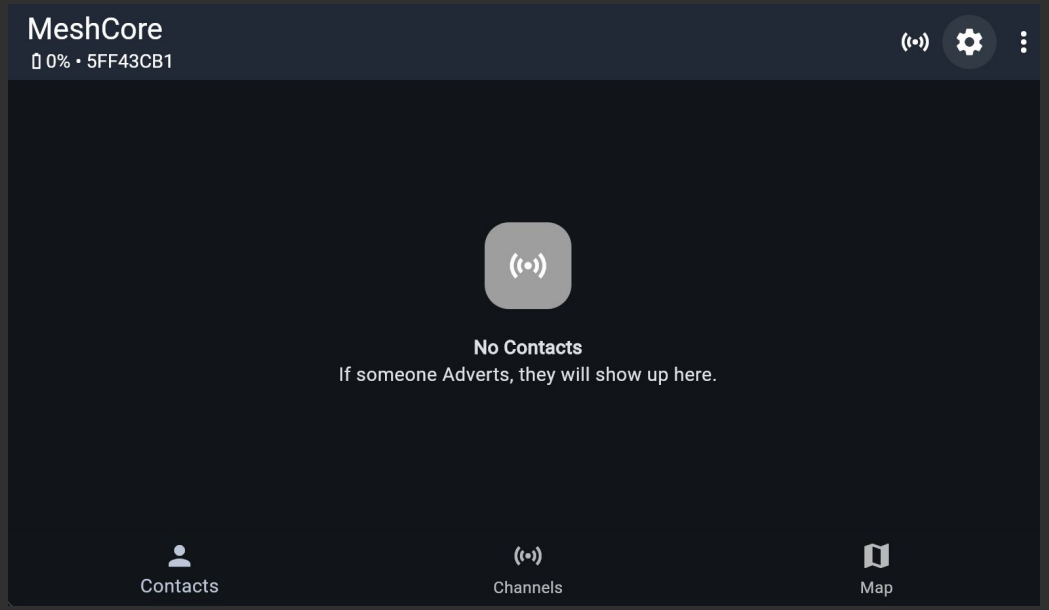
Connect to your device on your smartphone or https://app.meshcore.nz/



Pair Via Bluetooth & Open Settings

On https://app.meshcore.nz/ or the app, select your device and pair using the bluetooth pin. ****YOU MAY NEED TO FORGET THE DEVICE IN BLUETOOTH SETTINGS FOR THIS TO WORK****

Click the gear icon for settings

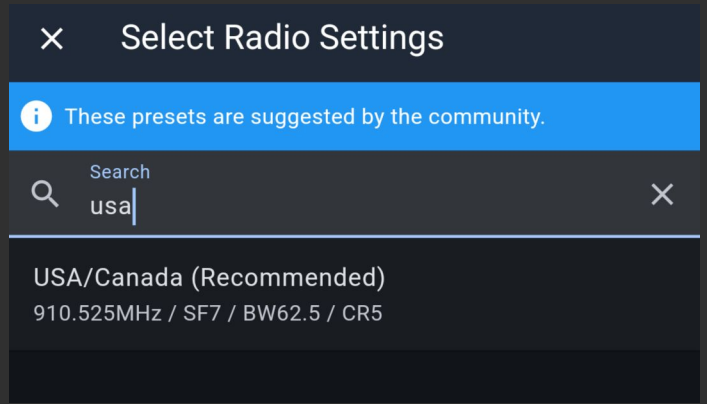


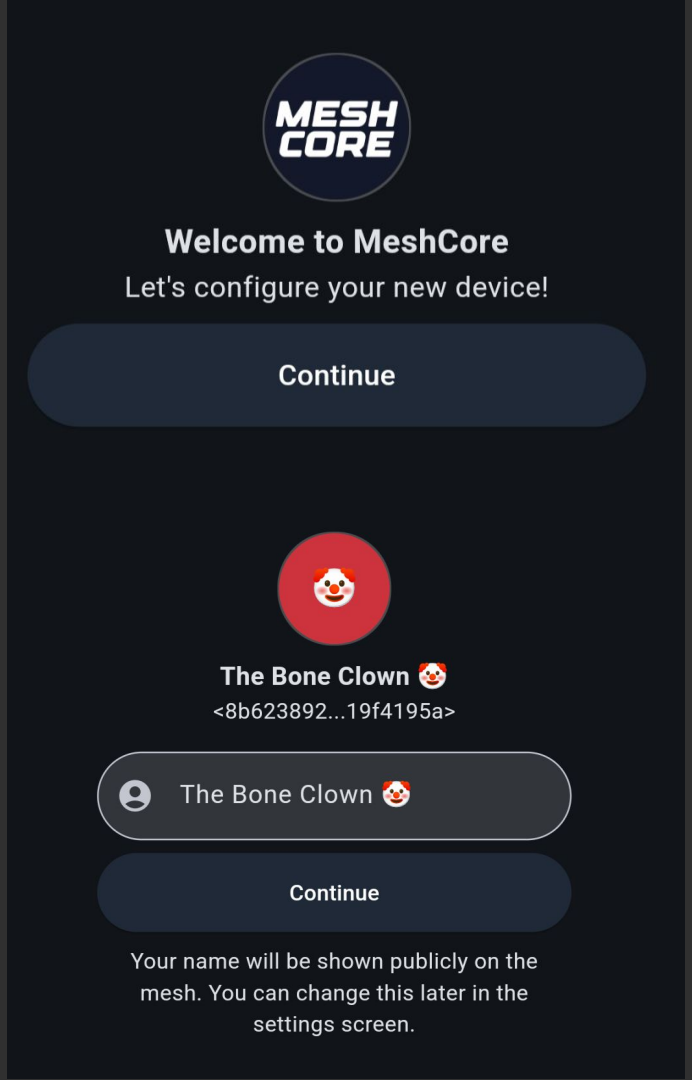
Save Frequency & Advertise

Set the frequency and save!

Next, hit “Advert - Flood Routed” to advertise your node to other devices in the area!

You’ll automatically add contacts you see.

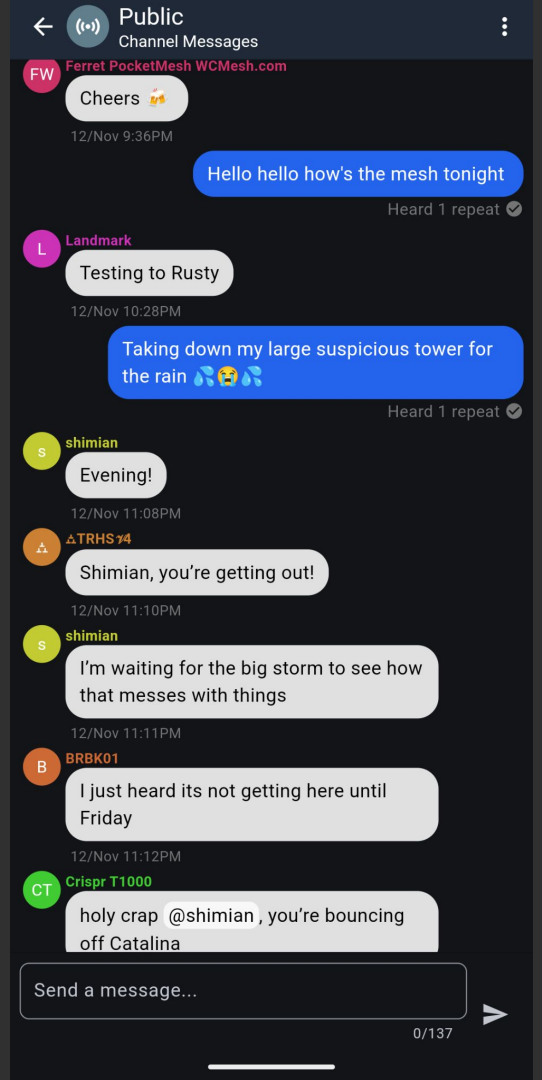




Try Chatting!

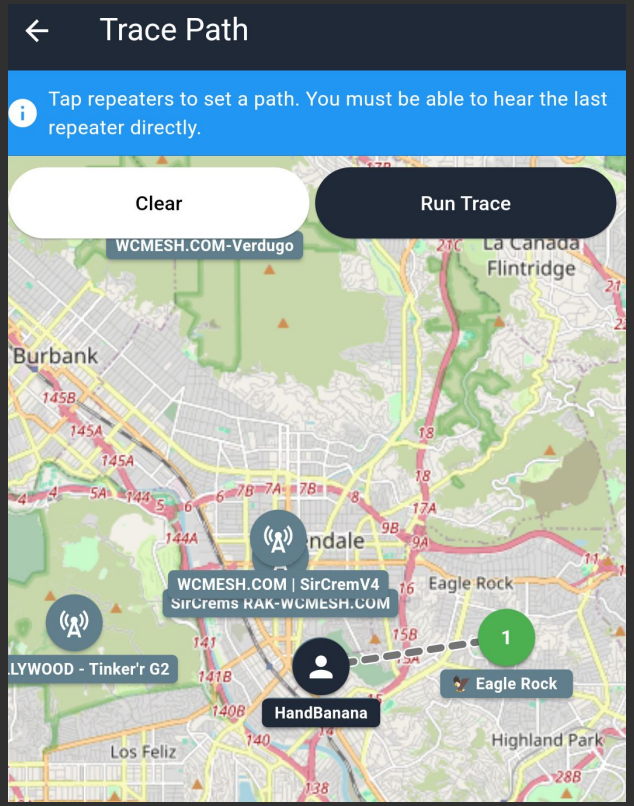
Try sending a message in the public chat! As long as you're on the same settings, it should work fast.

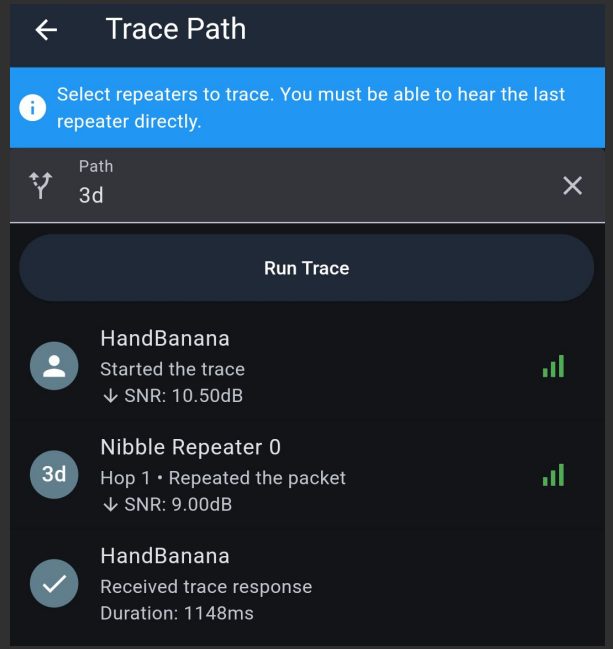
Users should be able to direct message you after seeing your advertisement.



Meshcore Tools To Explore: Trace Path

Trace a route between nodes using a map or text interface

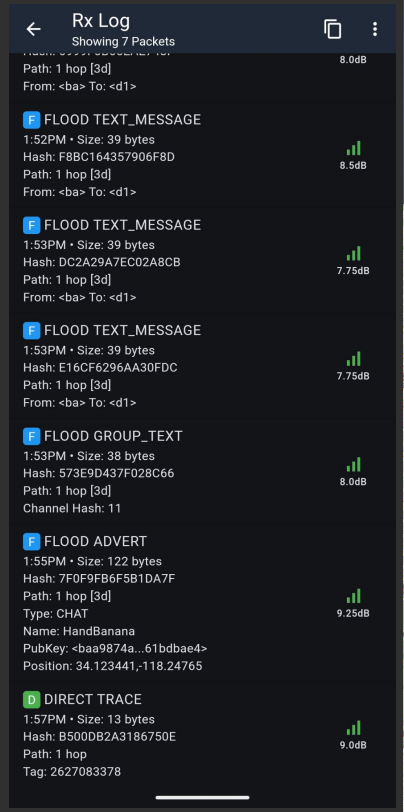


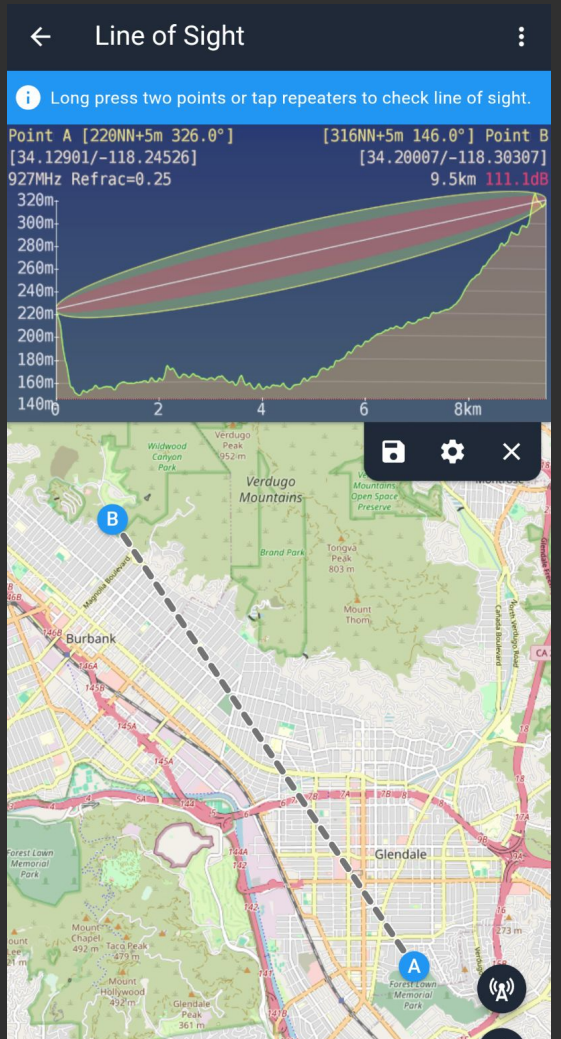


Meshcore Tools To Explore: Line of Sight

Check line of sight
between node locations

Also see all received
messages in RX log!





Meshcore Repeater



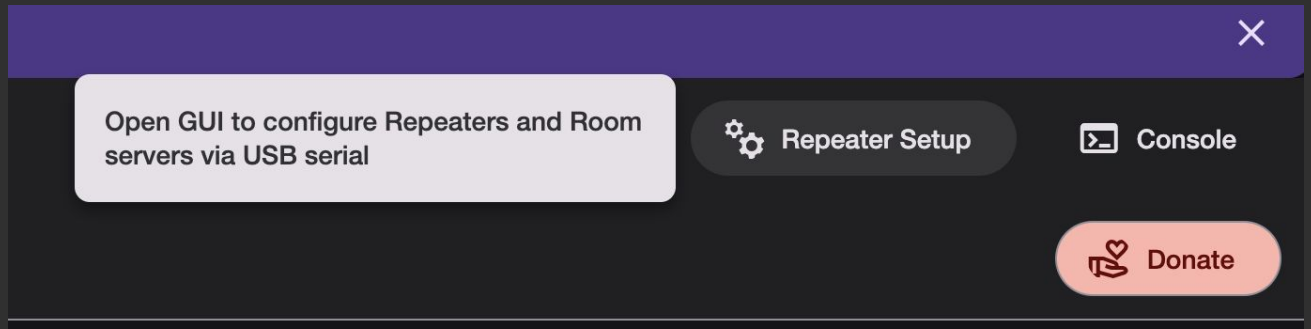
Want to flash a Meshcore Repeater?

Put your Nibble into boot mode and visit Scriptkitty.sh

Erase your device, select the Nibble Meshcore Repeater & flash

Unplug your device and plug it back in, then go to:

https://meshcore.io/flasher



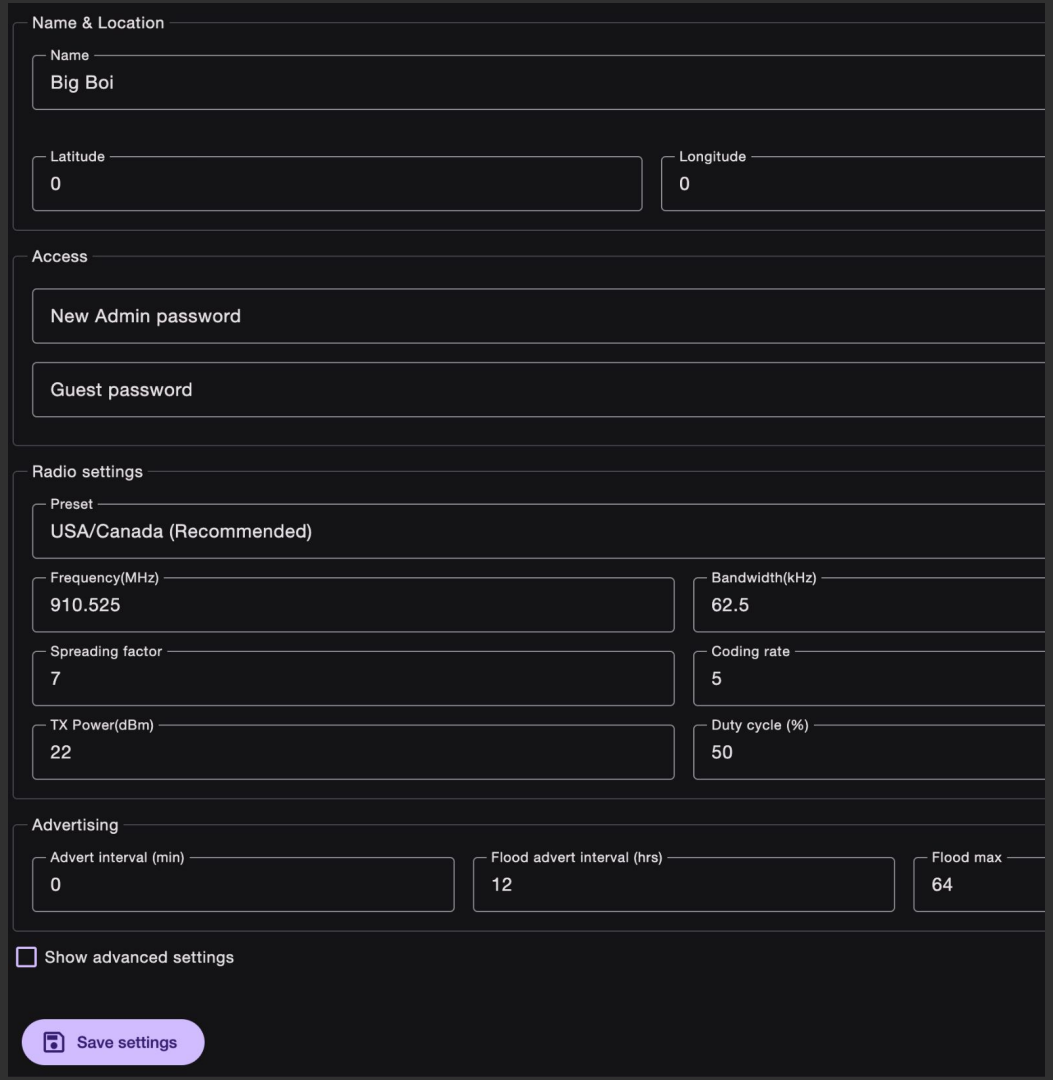
Set Up Repeater

Set the name and password

Change the radio settings to:

US Default

Save, and click **“Send Advert”**

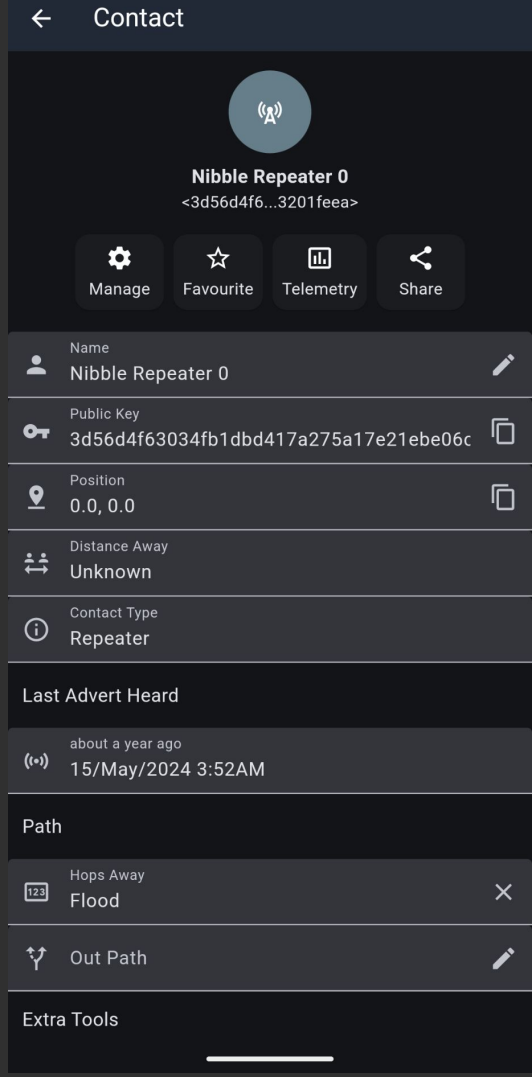


Configure Remotely Via Password

You can connect to your repeater via the password you set from another Meshtastic companion node.

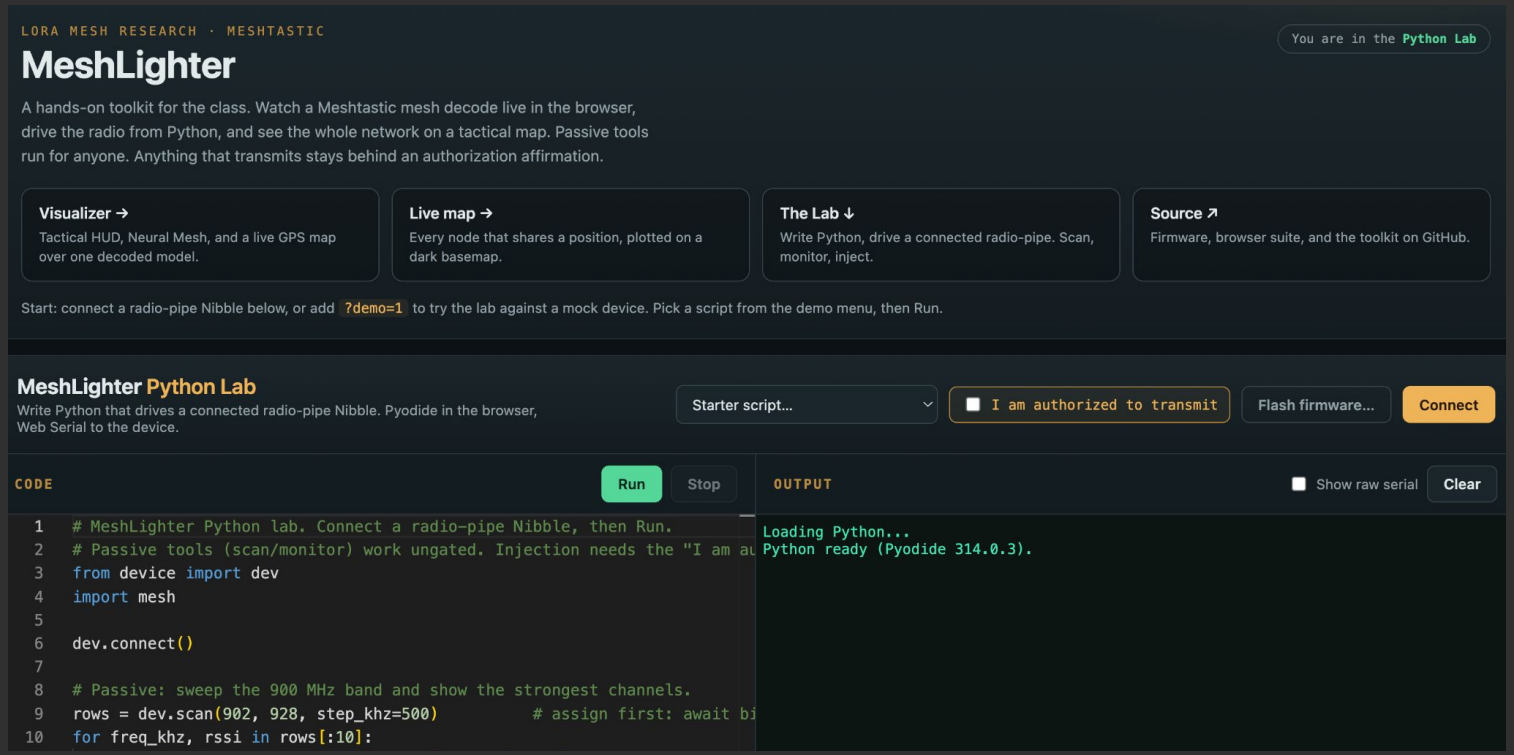
Find the repeater in your contacts and select **“Manage”**.

Enter the password you set and you’ll be able to access settings.



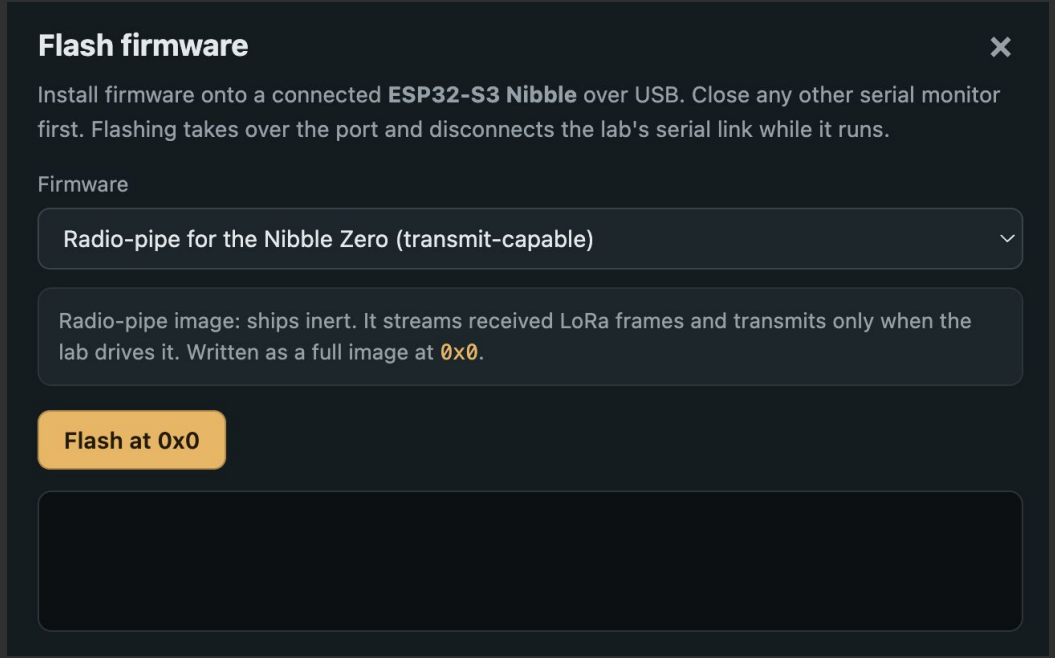
Final Lab: Hacking Meshtastic with Meshlighter.site

Meshlighter gives you raw LoRa radio access - Inject or listen



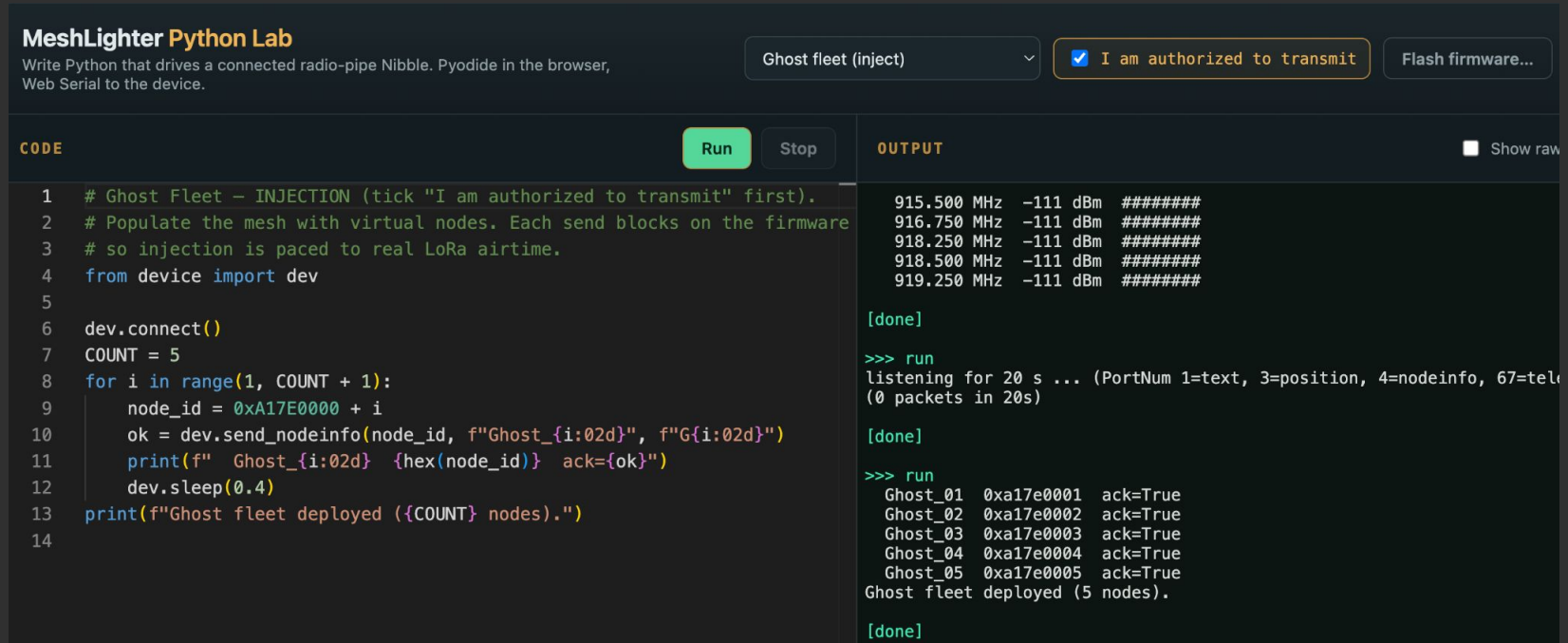
Final Lab: Hacking Meshtastic with Meshlighter.site

Go to www.meshlighter.site and flash the MeshLighter firmware to your board.



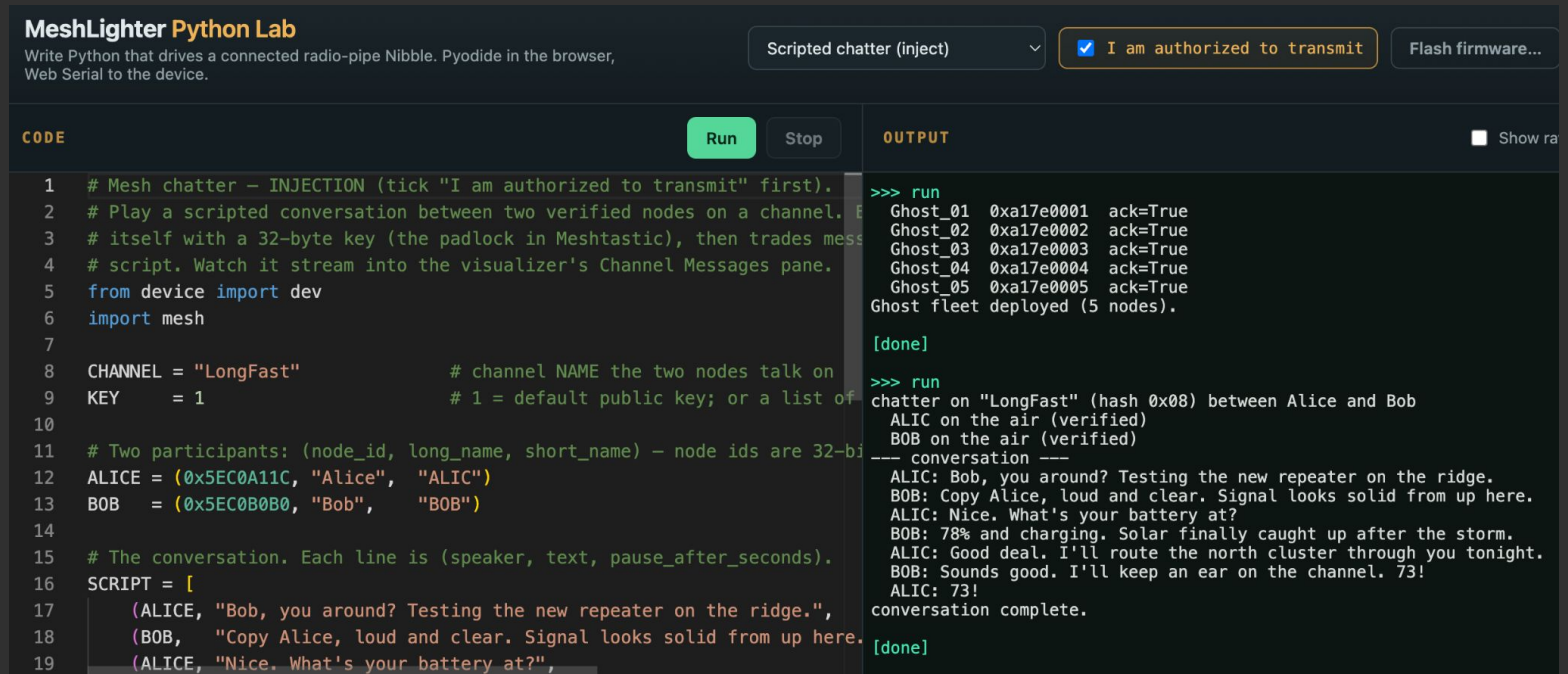
Hacking LoRa with Python in the Browser

In the Python labs, click connect, “I am authorized to transmit,” and then try out one of the demos, like injecting fake nodes.

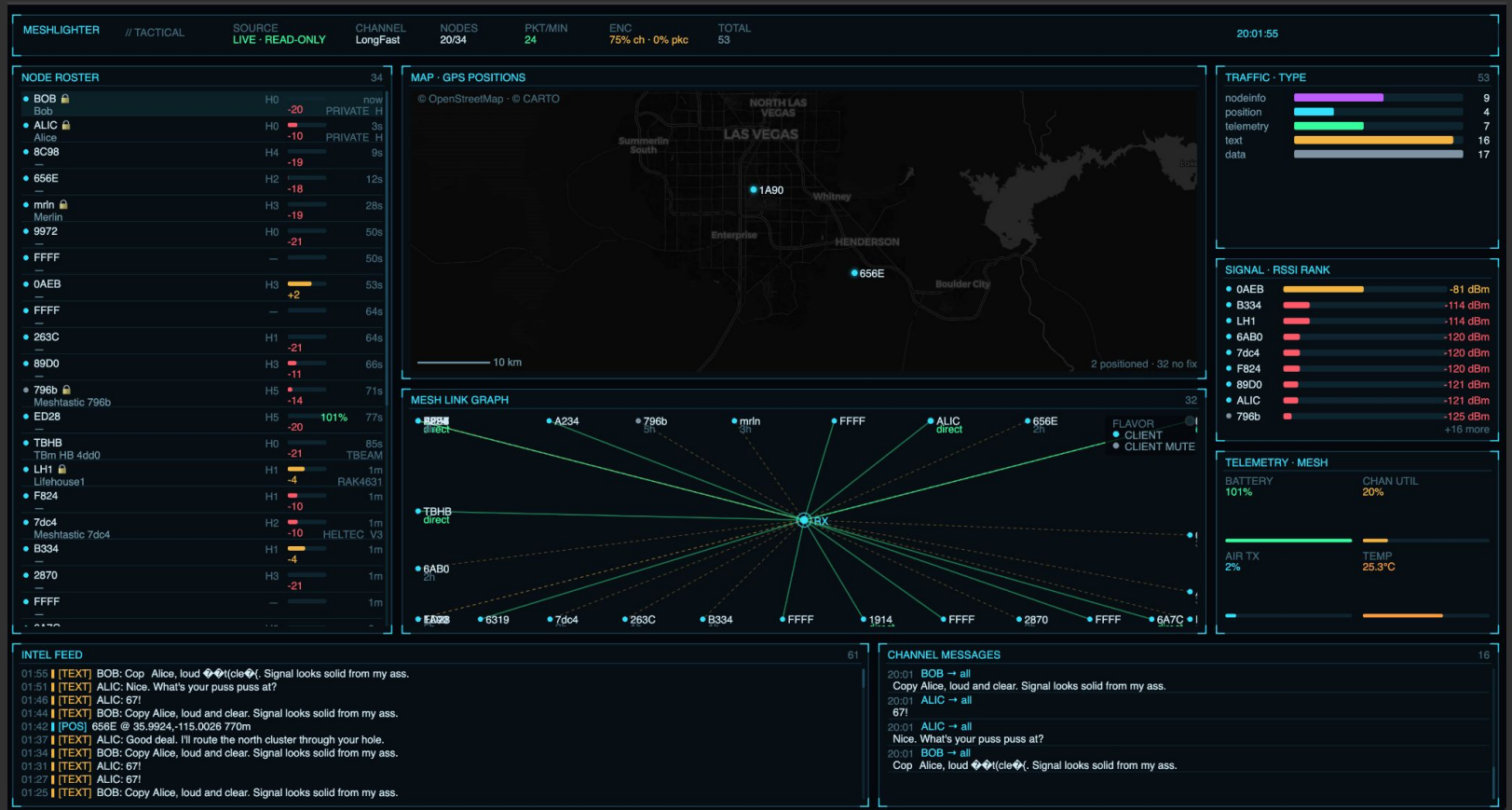


Script Fake Conversations Between Spoofed Nodes

The demos can spoof verified or unverified nodes, inject messages from nonexistant nodes, and create fake conversations



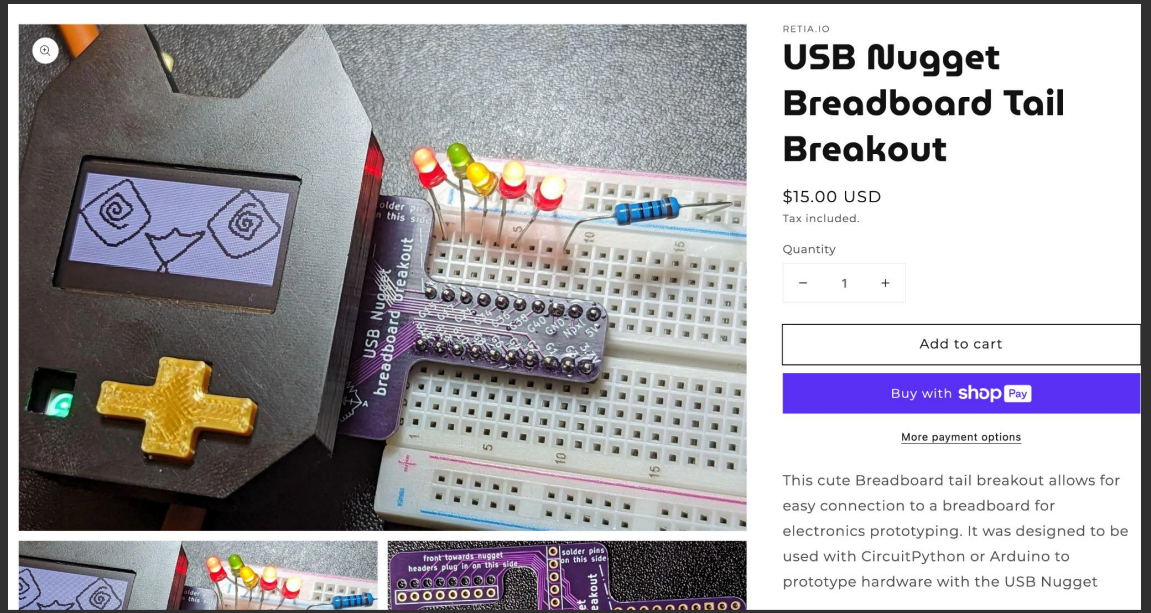
Signals Intelligence Dashboard



Buy a kit or make your own!

We have kits on Retia.io and discounts for instructors teaching classes!

- Nibbles
- Nuggets
- Add-ons
- Online classes



Keep in Touch

Want to learn more? You can find us here:

- Discord: https://discord.gg/rjVJbauAUX
- Store: Retia.io
- Nugget Flasher: Nugget.dev
- My work: www.hack.gay

